
Generalized Distribution-Free Semi-Supervised Learning with Risk Rewrite

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Abstract

Typical semi-supervised learning (SSL) methods rely on distributional assumptions, and their performance degrades when these are violated. While PNU learning, a risk rewriting method, offers a distribution-free alternative, it is restricted to binary classification and its variance optimality remains unclear. In this paper, we propose a generalized framework that constructs unbiased risk estimators using linear combinations of component risks, subsuming PNU learning and extending to multiclass classification. We derive the minimum achievable variance, demonstrating our estimator can attain lower variance than PNU in asymmetric loss scenarios. Furthermore, we establish a generalization bound directly linking this variance reduction to improved learning performance. Based on these theoretical insights, we introduce two practical SSL methods that empirically match or outperform existing approaches on binary and multiclass benchmarks.

1 INTRODUCTION

Although machine learning models typically require substantial labeled data to achieve good performance, acquiring such labels in real-world scenarios is often difficult and expensive. In contrast, unlabeled data is cheaper to acquire. Consequently, semi-supervised learning (SSL) has been extensively studied to leverage this unlabeled data when labeled data is scarce. Existing methods generally rely on specific distributional assumptions about the data or heuristics [Chapelle et al., 2006, Ouali et al., 2020, Amini et al., 2025]. Typical distributional assumptions include the *manifold assumption*, which posits that high-dimensional data lie on a lower-dimensional manifold, and the *cluster assumption*, which assumes that the decision boundary should pass

through low-density regions to separate data clusters [Ouali et al., 2020]. To encode these distributional assumptions into training a classifier, two main research directions have been studied.

The first is consistency regularization, which regularizes a classifier so that the prediction does not change significantly against the perturbation of unlabeled data. This improves the smoothness of the classifier along the data manifold. Early works were developed with graph-based regularization [Zhou et al., 2003, Belkin et al., 2006]. Recent methods based on deep models employ regularizers to enhance model robustness against input perturbations, data augmentation and dropout [Miyato et al., 2019, Laine and Aila, 2017, Tervainen and Valpola, 2017, Xie et al., 2020].

The second is Self-Training, which jointly trains a classifier and assigns labels (pseudo labels) to unlabeled data using current predictions. It pushes the decision boundary into low-density regions by discouraging low-confidence predictions [Ouali et al., 2020]. The pseudo-labeling process has traditionally been managed by fixed confidence thresholds [Yarowsky, 1995], and recent works have adjusted them with training progress [Zhang et al., 2021, Wang et al., 2023]. These methods are refined by multiple classifiers [Blum and Mitchell, 1998, Zhou and Li, 2005, Pham et al., 2021] to derive more accurate pseudo labels. Please refer to Amini et al. [2025], Ouali et al. [2020] for other SSL methods, which are generally based on the distributional assumptions mentioned above.

While these approaches have achieved remarkable success, their performance is often contingent on whether the underlying distributional assumptions hold for the target dataset. If these assumptions are violated, the inductive bias introduced by the unlabeled data can behave adversely, potentially degrading performance below that of supervised learning [Cozman et al., 2003, Li and Zhou, 2011, Krijthe and Loog, 2017, Wang et al., 2022, Arazo et al., 2020].

Unbiased Risk Estimators and Risk Rewriting. A distinct line of research focuses on utilizing unlabeled data

for risk evaluation without relying on restrictive distributional assumptions. This is achieved through *risk rewriting* [Sugiyama et al., 2022], which reformulates the standard expected risk into an unbiased estimator computable directly from the available data distributions (e.g., combinations of labeled and unlabeled data). For instance, Positive-Unlabeled (PU) learning trains a classifier using only positive and unlabeled data by constructing such a rewritten risk estimator [du Plessis et al., 2014]. Sakai et al. [2017] extended this to the semi-supervised setting by proposing PNU learning, which combines PN (supervised), PU, and NU (Negative-Unlabeled) risks. They theoretically demonstrated that PNU learning can reduce the variance of risk estimators compared to standard supervised learning. This approach is compelling as it is theoretically supported and applicable to any model without requiring strong distributional assumptions (such as the cluster or manifold assumptions).

Limitations and Open Problems. Despite its theoretical advantages, the PNU learning framework proposed by Sakai et al. [2017] leaves several open problems. First, their method is limited to binary classification. Second, it is unclear whether PNU is variance-optimal among broader classes of unbiased risk-rewriting estimators. In this paper, we address these limitations by generalizing the method to a linear combination set of risks S_{lin} and proposing new SSL methods to minimize the variance of the risk estimator.

Our contributions are summarized as follows:

- We propose a generalized framework for risk-rewriting SSL that subsumes PNU risk and extends naturally to the multiclass setting. As discussed in Section 7, our framework has a potential impact on a broad range of SSL and weakly supervised learning problems.
- We derive the theoretical minimum variance achievable by the set S_{lin} and show it can have a smaller minimum variance than PNU risk in the general (asymmetric) loss case. In the symmetric loss case, we show PNU risk can achieve optimal minimum variance in S_{lin} .
- We propose two novel SSL methods: (1) an iterative optimization method and (2) a data-free method under an equal covariance assumption, and empirically demonstrate that they consistently outperform or match existing SSL methods.

2 PROBLEM SETTING

We consider a multiclass classification problem where the input space is $\mathcal{X} \subset \mathbb{R}^d$ and the output label space is $\mathcal{Y} = \{1, \dots, k\}$. We assume the data is generated from a joint distribution $p(x, y)$ over $\mathcal{X} \times \mathcal{Y}$. Let $p_i(x) = p(x \mid y = i)$ denote the class-conditional density for class i , and let $\theta_i = P(y = i)$ be the class prior probability, such that $\sum_{i=1}^k \theta_i = 1$. The distribution of unlabeled data is given

by the mixture model: $p(x) = \sum_{i=1}^k \theta_i p_i(x)$. In the semi-supervised setting, we are provided with labeled datasets $\mathcal{X}_i = \{x_j^i\}_{j=1}^{n_i} \stackrel{i.i.d.}{\sim} p_i(x)$ for each class $i \in \mathcal{Y}$, and an unlabeled dataset $\mathcal{X}_U = \{x_j^U\}_{j=1}^{n_U} \stackrel{i.i.d.}{\sim} p(x)$.

Risk Definitions. Let $g : \mathcal{X} \rightarrow \mathbb{R}^k$ be a decision function (e.g., a neural network) and $l : \mathbb{R}^k \times \mathcal{Y} \rightarrow \mathbb{R}$ be a loss function. Assume that g belongs to a function class $\mathcal{G}$. The goal of standard supervised learning is to minimize the true risk $R(g) := \mathbb{E}_{(x,y) \sim p(x,y)} [l(g(x), y)]$.

To investigate unbiased risk estimators, we decompose the risk into components based on the data distribution and the label used for loss evaluation. We define component risks $R_{ij}(g)$ as:

$$R_{ij}(g) := \mathbb{E}_{x \sim p_i} [l(g(x), j)],$$

which is the expected loss of predictor g calculated over the distribution of class i but evaluated with respect to a fixed label j . Similarly, we define the risk over the unlabeled distribution with respect to label j as:

$$R_{Uj}(g) := \mathbb{E}_{x \sim p(x)} [l(g(x), j)].$$

Using these components, the standard supervised risk can be expressed as $R(g) = \sum_{i=1}^k \theta_i R_{ii}(g)$.

2.1 REVISITING BINARY RISK REWRITING: PU, NU, AND PNU RISKS

Although the standard supervised risk is written as $R(g) = \sum_{i=1}^k \theta_i R_{ii}(g)$, it is not the only expression. In binary classification, several different formulations have been proposed for $R(g)$ with unlabeled data. Consider the case where $k = 2$, with $y = 1$ and $y = 2$ representing the positive and negative classes, respectively. Then the standard supervised risk (PN risk) is $R_{PN}(g) := \theta_1 R_{11}(g) + \theta_2 R_{22}(g)$.

PU Risk. Positive-Unlabeled (PU) learning [du Plessis et al., 2014] addresses scenarios where negative labels are unavailable. By exploiting the density equality $\theta_2 p_2(x) = p(x) - \theta_1 p_1(x)$, one can rewrite the risk on the negative class as $\theta_2 R_{22}(g) = R_{U2}(g) - \theta_1 R_{12}(g)$. This yields the PU risk:

$$R_{PU}(g) := \theta_1 R_{11}(g) - \theta_1 R_{12}(g) + R_{U2}(g).$$

This risk is equivalent to the PN risk, but formulated without requiring the negative distribution.

NU Risk. By applying a symmetric transformation (substituting the positive component via $\theta_1 p_1(x) = p(x) - \theta_2 p_2(x)$), we obtain the Negative-Unlabeled (NU) risk:

$$R_{NU}(g) := \theta_2 R_{22}(g) - \theta_2 R_{21}(g) + R_{U1}(g).$$

PNU Risk. Under certain conditions and as $n_U \rightarrow \infty$, PU and NU learning outperform PN learning [Niu et al.,

2016]. To exploit this advantage further, Sakai et al. [2017] proposed PNU learning¹, which considers the following risk that linearly combines PU, NU, and PN risks with a parameter $\eta \in [-1, 1]$:

$$R_{\text{PNU}}^\eta(g) := \begin{cases} R_{\text{PNPU}}^\eta(g) & (\eta \geq 0) \\ R_{\text{PNNU}}^{|\eta|}(g) & (\eta < 0) \end{cases}$$

where

$$\begin{aligned} R_{\text{PNPU}}^\eta(g) &:= (1 - \eta)R_{\text{PN}}(g) + \eta R_{\text{PU}}(g), \\ R_{\text{PNNU}}^\eta(g) &:= (1 - \eta)R_{\text{PN}}(g) + \eta R_{\text{NU}}(g). \end{aligned}$$

The estimation of PNU risk with empirical distributions has a smaller variance than that of PN risk, which leads to more efficient SSL. This method has several advantages: (i) no requirement of distributional assumptions unlike other SSL methods, (ii) compatibility with any loss and classification model, and (iii) minimal additional computational costs. Unlike augmentation-heavy or teacher-student SSL methods [Zhang et al., 2021, Pham et al., 2021], our methods do not require augmentation passes or extra networks.

Despite these advantages, PNU learning is limited to binary classification. How to extend it to multiclass classification is not obvious. In addition, it is not clear if we can construct a risk with a smaller estimation variance than that of the PNU risk.

3 OUR PROPOSED FRAMEWORK: GENERALIZED RISK REWRITE

In this section, we propose a generalized framework for semi-supervised learning based on risk rewriting. We introduce a broad class of unbiased risk estimators formed by linear combinations of component risks and analyze their variance properties.

3.1 GENERALIZED UNBIASED ESTIMATORS VIA LINEAR COMBINATIONS

We define the set of linear combinations of risk functionals, S_{lin} , as

$$S_{\text{lin}} := \left\{ R_{\text{lin}}^{\{a_{ij}\}, \{b_j\}}(\cdot) := \sum_{i,j=1}^k a_{ij} R_{ij}(\cdot) + \sum_{j=1}^k b_j R_{Uj}(\cdot) \mid a_{ij}, b_j \in \mathbb{R}, \quad \forall g \in \mathcal{G}, R_{\text{lin}}^{\{a_{ij}\}, \{b_j\}}(g) = R(g) \right\}.$$

The set is defined as linear combinations of component risks and unlabeled risks, which are equal to $R(g)$ for all $g \in$

¹They also proposed PUNU risk, but we omit it since it was shown to be inferior to PNU risk.

$\mathcal{G}$. It is important to note that this formulation generalizes existing risks such as PU, NU, and PNU risks. We can build unbiased estimators for each $R_{\text{lin}}^{\{a_{ij}\}, \{b_j\}}(\cdot) \in S_{\text{lin}}$ by using empirical distributions. By optimizing the parameters a_{ij} and b_j over the full set S_{lin} , we can derive a risk that has a lower estimation variance than the existing methods.

3.2 CHARACTERIZATION UNDER LINEAR INDEPENDENCE

The number of parameters that define the risks in S_{lin} can be reduced under the following mild assumption.

Assumption 1 (Linear Independence). *The risk components are linearly independent in the sense that:*

$$\sum_{i,j=1}^k c_{ij} R_{ij}(g) = 0 \quad (\forall g \in \mathcal{G}) \implies \forall i, j, c_{ij} = 0.$$

This assumption is satisfied when $\mathcal{G}$ is reasonably large and the loss function is not constrained by linear relations. In particular, it cannot be met by symmetric loss functions (such as the 0-1 loss), because $\sum_{j=1}^k l(g(x), j)$ is constant and the risk components are linearly dependent. The symmetric-loss case is discussed in Section 4.

Under Assumption 1, we can characterize the parameters that satisfy the constraint $\forall g \in \mathcal{G}, R_{\text{lin}}^{\{a_{ij}\}, \{b_j\}}(g) = R(g)$.

Theorem 1 (Parametrization of S_{lin}). *Under Assumption 1, S_{lin} can be rewritten as*

$$S_{\text{lin}} = \left\{ R_{\text{lin}}^{\mathbf{a}}(\cdot) := \sum_{i=1}^k \theta_i R_{ii}(\cdot) + \sum_{i,j=1}^k \theta_i \left(\frac{a_j}{\theta_j} - 1 \right) R_{ij}(\cdot) - \sum_{j=1}^k \left(\frac{a_j}{\theta_j} - 1 \right) R_{Uj}(\cdot) \mid \mathbf{a} \in \mathbb{R}^k \right\}.$$

This theorem implies that each risk in S_{lin} is parametrized by a k -dimensional vector.

3.3 MINIMUM VARIANCE CALCULATION

In the following, we use a fixed g and often omit the argument of the risk term, denoting $R_{ij}(g)$ as R_{ij} when the context is clear. Let $\hat{R}_{\text{lin}}^{\mathbf{a}}$ and $\hat{R}_{ij}$ denote the empirical estimators that are defined by replacing true distributions with empirical distributions. We consider the asymptotic regime where the number of unlabeled samples $n_U \rightarrow \infty$.

Let $C_m \in \mathbb{R}^{k \times k}$ be the covariance matrix for class m , defined as $(C_m)_{ij} := \text{Cov}_{x \sim p_m}[l(g(x), i), l(g(x), j)]$. The covariance of the empirical risk estimates is given by $\text{Cov}[\hat{R}_{mi}, \hat{R}_{mj}] = \frac{1}{n_m} (C_m)_{ij}$. We define the diagonal matrix $Q = \text{diag}(\theta_1^{-1}, \dots, \theta_k^{-1})$ and the shift vectors

$\mathbf{d}_m = \mathbf{1} - \mathbf{e}_m$, where $\mathbf{1}$ is the all-ones vector and $\mathbf{e}_m$ is the standard basis vector. We now derive the variance of the estimator $\hat{R}_{lin}^a$.

Theorem 2 (Variance of $\hat{R}_{lin}^a$).

$$\begin{aligned} \text{Var}[\hat{R}_{lin}^a] &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} (Q\mathbf{a} - \mathbf{d}_m)^T C_m (Q\mathbf{a} - \mathbf{d}_m) \\ &= \mathbf{a}^T A \mathbf{a} - 2\mathbf{b}^T \mathbf{a} + c, \end{aligned}$$

where we define the system matrices as

$$\begin{aligned} A &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} Q C_m Q, \quad \mathbf{b} = \sum_{m=1}^k \frac{\theta_m^2}{n_m} Q C_m \mathbf{d}_m, \\ c &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} \mathbf{d}_m^T C_m \mathbf{d}_m. \end{aligned}$$

Since A is a sum of positive semi-definite matrices, the variance is a convex function of $\mathbf{a}$. Define $f(\mathbf{a}) := \text{Var}[\hat{R}_{lin}^a]$. The optimal parameter vector $\mathbf{a}^*$ that minimizes the variance satisfies the first-order condition:

$$\nabla_{\mathbf{a}} f(\mathbf{a}) = 2A\mathbf{a} - 2\mathbf{b} = 0 \implies A\mathbf{a}^* = \mathbf{b}.$$

If A is invertible, the minimum variance is given by $f(\mathbf{a}^*) = c - \mathbf{b}^T A^{-1} \mathbf{b}$. For simplicity, we define the total weighted covariance matrix S and the weighted covariance vector $\mathbf{u}$:

$$S := \sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m, \quad \mathbf{u} := \sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m \mathbf{e}_m.$$

Then, we derive the following theorem:

Theorem 3 (Minimum Variance of $\hat{R}_{lin}^a$). Assume $n_U \rightarrow \infty$ and that S is invertible. The minimum variance achievable by the linear risk estimator $\hat{R}_{lin}^a$ is given by

$$\min_{\mathbf{a}} \text{Var}[\hat{R}_{lin}^a] = \sum_{m=1}^k \frac{\theta_m^2}{n_m} (C_m)_{mm} - \mathbf{u}^T S^{-1} \mathbf{u}.$$

The first term represents the pointwise variance of the supervised estimator for the same fixed g , and the second term $\mathbf{u}^T S^{-1} \mathbf{u}$ (which is non-negative as S is positive semi-definite) represents the variance reduction from the proposed semi-supervised method.

3.4 MAGNITUDE OF VARIANCE REDUCTION

The variance reduction $\mathbf{u}^T S^{-1} \mathbf{u}$ depends on the sample sizes, class priors and covariance matrices. To analyze when the reduction is large, we assume $\forall m \in [k], C_m = C$ where $(C)_{ii} = \rho_1 > 0$ and $(C)_{ij} = \rho_2 < 0$ for $i \neq j$, since $(C_m)_{ij}$ for $i \neq j$ tends to be negative for usual loss functions.

Theorem 4 (Variance reduction). Assume C is invertible and $\forall m \in [k], C_m = C$. Then,

$$\begin{aligned} \mathbf{u}^T S^{-1} \mathbf{u} &= (\rho_1 - \rho_2) \frac{\sum_{m=1}^k w_m^2}{W} + \rho_2 W \\ &\leq \rho_1 \max_{m \in [k]} w_m + \rho_2 \left(W - \max_{m \in [k]} w_m \right), \end{aligned}$$

where $w_m := \frac{\theta_m^2}{n_m}$ and $W := \sum_{m=1}^k w_m$.

Considering the upper bound of $\mathbf{u}^T S^{-1} \mathbf{u}$, the variance reduction is large when $\max_{m \in [k]} w_m$ is large and $W - \max_{m \in [k]} w_m$ is small, which means θ_i and n_i should be disproportionate, i.e., there exists class imbalance in labeled samples.

3.5 COMPARISON OF OPTIMAL VARIANCES WITH PNU LEARNING

We compare the minimum variance achievable by our proposed method with that of PNU learning in binary classification. This analysis is different from the section 3.4, which studies the absolute variance reduction relative to supervised learning. Here, we instead quantify the relative advantage of the proposed estimator over PNU learning. For simplicity, we compare our proposed estimator against the PNPU and PNNU estimators.

Theorem 5 (Variance comparison). Assume $k = 2$ and $n_U \rightarrow \infty$. Let $\eta_{PU}^* := \arg\min_{\eta \in \mathbb{R}} \text{Var}(\hat{R}_{PNPU}^\eta)$ and $\eta_{NU}^* := \arg\min_{\eta \in \mathbb{R}} \text{Var}(\hat{R}_{PNNU}^\eta)$ be the optimal parameters for the PNPU and PNNU estimators respectively. Then,

$$\text{Var}(\hat{R}_{lin}^{a^*}) \leq \min \left(\text{Var}(\hat{R}_{PNPU}^{\eta_{PU}^*}), \text{Var}(\hat{R}_{PNNU}^{\eta_{NU}^*}) \right). \quad (1)$$

Furthermore, under the same assumption as Theorem 4,

$$\begin{aligned} &\min \left(\text{Var}(\hat{R}_{PNPU}^{\eta_{PU}^*}), \text{Var}(\hat{R}_{PNNU}^{\eta_{NU}^*}) \right) - \text{Var}(\hat{R}_{lin}^{a^*}) \\ &= \min \left(\frac{w_1^2(\rho_1^2 - \rho_2^2)}{(w_1 + w_2)\rho_1}, \frac{w_2^2(\rho_1^2 - \rho_2^2)}{(w_1 + w_2)\rho_1} \right) \end{aligned} \quad (2)$$

Eq. (1) implies our proposed estimator has a minimum variance less than or equal to that of PNPU and PNNU estimators, which is rather trivial since the set of $\hat{R}_{lin}^a$ subsumes the sets of $\hat{R}_{PNPU}^\eta$ and $\hat{R}_{PNNU}^\eta$. Eq. (2) implies the gap in minimum variances becomes small when $w_1 \gg w_2$ or $w_1 \ll w_2$, which means our method is more advantageous when θ_i and n_i are proportional, i.e., the labeled data is balanced.

4 SYMMETRIC LOSS CASE

The preceding sections analyzed the properties of S_{lin} under Assumption 1. For symmetric loss functions such as the 0-1 loss, Assumption 1 does not hold, and we need a modified analysis, as described in this section.

4.1 PROBLEM SETUP UNDER SYMMETRIC LOSS

We assume the loss function $l(\cdot, \cdot)$ satisfies the symmetric condition:

Definition 1 (Symmetric Loss). *A loss function $l(\cdot, \cdot)$ is said to be **symmetric** if, for any input x and any prediction vector $g(x)$, the sum of losses over all possible class labels $j \in \{1, \dots, k\}$ is constant:*

$$\sum_{j=1}^k l(g(x), j) = \alpha, \quad \forall x \in \mathcal{X}, \forall g,$$

where α is a constant.

Common examples include the 0-1 loss and ramp loss [Charoenphakdee et al., 2019]. This constraint implies a deterministic linear dependency among the risk components: $\sum_{j=1}^k R_{ij}(g) = \alpha$. Consequently, the linear independence assumption (Assumption 1) does not hold, and we need a different reparametrization for S_{lin} .

4.2 REPARAMETRIZATION OF S_{lin} WITH SYMMETRIC LOSS

To analyze the class of risks, we introduce a modified assumption similar to linear independence.

Assumption 2.

$$\sum_i^k \sum_j^{k-1} c_{ij} R_{ij}(g) + c = 0 (\forall g \in \mathcal{G})$$

$$\Rightarrow \forall i \in [k], j \in [k-1], c_{ij} = 0 \text{ and } c = 0.$$

Under this assumption, we can characterize S_{lin} similarly to Theorem 1.

Theorem 6. *Under Assumption 2, S_{lin} with a symmetric loss can be rewritten as*

$$S_{lin} = \left\{ R_{lin}^a(\cdot) := \sum_{i=1}^k \theta_i R_{ii}(\cdot) + \sum_{i=1}^k \sum_{j=1}^{k-1} \theta_i \left(\frac{a_j}{\theta_j} - 1 \right) R_{ij}(\cdot) - \sum_{j=1}^{k-1} \left(\frac{a_j}{\theta_j} - 1 \right) R_{Uj}(\cdot) \mid \mathbf{a} \in \mathbb{R}^{k-1} \right\}.$$

Thus, S_{lin} with a symmetric loss is parametrized by the $k-1$ parameters, $\mathbf{a}_i, i \in [k-1]$.

4.3 VARIANCE OF $\hat{R}_{lin}^a$ WITH SYMMETRIC LOSS

Since the symmetric loss constraint implies $R_{mk} = \alpha - \sum_{j=1}^{k-1} R_{mj}$, the k -th row and column of the covariance matrix C_m are entirely determined by the covariance of the first $k-1$ risk terms. Thus, the variance of $\hat{R}_{lin}^a$ is expressed with truncated matrices of C_m . Let $\check{C}_m$ be the top-left $(k-1) \times (k-1)$ submatrix of C_m .

Theorem 7 (Variance of $\hat{R}_{lin}^a$ with symmetric loss). *With a symmetric loss function $l(\cdot, \cdot)$,*

$$\text{Var} \left[\hat{R}_{lin}^a \right] = \sum_{m=1}^k \frac{\theta_m^2}{n_m} (\check{Q} \mathbf{a} - \check{\mathbf{d}}_m)^T \check{C}_m (\check{Q} \mathbf{a} - \check{\mathbf{d}}_m)$$

$$\text{where } \check{Q} = \text{diag}\left(\frac{1}{\theta_1}, \dots, \frac{1}{\theta_{k-1}}\right) \text{ and } \check{\mathbf{d}}_m = \begin{cases} \mathbf{1} - \mathbf{e}_m & \text{if } m < k \\ 2\mathbf{1} & \text{if } m = k \end{cases}.$$

To derive the minimum variance, we define $\check{S} := \sum_{m=1}^k \frac{\theta_m^2}{n_m} \check{C}_m$, $\check{\mathbf{u}} := \sum_{m=1}^{k-1} \frac{\theta_m^2}{n_m} \check{C}_m \mathbf{e}_m - \frac{\theta_k^2}{n_k} \check{C}_k \mathbf{1}$. The minimum variance is derived similarly to Theorem 3.

Theorem 8 (Minimum Variance with Symmetric Loss). *Assume $\check{S}$ is invertible. The minimum variance achievable by the linear risk estimator $\hat{R}_{lin}^a$ under symmetric loss is given by*

$$\min_{\mathbf{a}} \text{Var} \left[\hat{R}_{lin}^a \right] = \sum_{m=1}^k \frac{\theta_m^2}{n_m} (C_m)_{mm} - \check{\mathbf{u}}^T \check{S}^{-1} \check{\mathbf{u}}.$$

4.4 COMPARISON IN BINARY CLASSIFICATION WITH SYMMETRIC LOSS

We compare the proposed method, PNPU and PNNU learning in binary classification ($k=2$) under the symmetric loss assumption. We demonstrate that in this restricted setting, these methods are mathematically equivalent.

Theorem 9 (Equivalence of risk estimators). *Consider the binary classification setting ($k=2$) with a symmetric loss function. Let $S_{PNPU} := \{R_{PNPU}^\eta \mid \eta \in \mathbb{R}\}$ and $S_{PNNU} := \{R_{PNNU}^\eta \mid \eta \in \mathbb{R}\}$ denote the sets of PNPU and PNNU risks. Then,*

$$S_{lin} = S_{PNPU} = S_{PNNU}.$$

Therefore, they achieve the same minimum variance, which implies that, PNU is already variance-optimal within the class of linear unbiased risk rewritings in this setting.

5 RELATIONSHIP BETWEEN SEMI-SUPERVISED LEARNING AND VARIANCE REDUCTION

The analysis in the previous sections focused on the point-wise variance of the unbiased risk estimator for a fixed classifier g . The goal of semi-supervised learning is to identify the optimal predictor $\hat{g}$ using the empirical risk. In this section, we establish the theoretical connection between reducing the variance of the risk estimator and improving the generalization bound of the learned classifier.

5.1 GENERALIZATION BOUND WITH REDUCED VARIANCE

We consider $n_U \rightarrow \infty$ to analyze the generalization performance. Let $\hat{g} = \operatorname{argmin}_{g \in \mathcal{G}} \hat{R}_{lin}^a(g)$ be the empirical risk minimizer using our proposed estimator, and let $g^* = \operatorname{argmin}_{g \in \mathcal{G}} R(g)$ be the true risk minimizer.

Theorem 10 (Generalization Bound for $\hat{R}_{lin}^a$). *Assume $n_U \rightarrow \infty$ and that the loss function $l(z, y)$ is L -Lipschitz continuous with respect to the prediction $z \in \mathbb{R}^k$, and bounded such that $0 \leq l(\cdot, \cdot) \leq c_l$. Assume $\max_i \left| \frac{a_i}{\theta_i} - 1 \right| \leq c_a$. Let $\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_\infty)$ be the ν -covering number of $\mathcal{G}$ with respect to the L_∞ norm [Wainwright, 2019]. Define the maximum variance on the ν -cover C_ν as $\sigma_{\max}^2(\mathbf{a}, \nu) := \max_{g' \in C_\nu} \operatorname{Var}[\hat{R}_{lin}^a(g')]$. Then, for any $\nu > 0$, with probability at least $1 - \delta$, the excess risk of $\hat{g}$ satisfies:*

$$R(\hat{g}) - R(g^*) \leq 4L_a\nu + \sqrt{8\sigma_{\max}^2(\mathbf{a}, \nu) \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_\infty)}{\delta} \right)} + \frac{4}{3}B_a \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_\infty)}{\delta} \right),$$

where $B_a := kc_l(1 + c_a)$ and $L_a := L(1 + kc_a)$.

Let $n = \min_m(n_m)$. As $n \rightarrow \infty$, the maximum variance $\sigma_{\max}^2(\mathbf{a}, \nu)$ scales as $\mathcal{O}(1/n)$, which implies that the second term of the generalization bound is $\mathcal{O}(1/\sqrt{n})$. Furthermore, since the third term is proportional to $\max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right)$, it scales as $\mathcal{O}(1/n)$. Considering only the dominant $\mathcal{O}(1/\sqrt{n})$ term and assuming ν is small, this theorem implies that the generalization error is small when the maximum variance $\sigma_{\max}^2(\mathbf{a}, \nu)$ is small. Thus, a good choice of $\mathbf{a}$ improves the SSL performance.

5.2 PRACTICAL IMPLEMENTATION OF SSL

Based on the relationship established above, we propose two SSL methods to learn the classifier $\hat{g}$.

1. Iterative optimization method Since the optimal parameter $\mathbf{a}$ that minimizes the bound in Theorem 10 is unknown in practice, we propose empirical iterative optimization of g and $\mathbf{a}$. The algorithm is described in Algorithm 1. It optimizes $\hat{g}$ with $\hat{R}_{lin}^a$, calculating $\mathbf{a}$ every m epochs to minimize the variance with the current $\hat{g}$ using validation data.

2. Data-free method under equal covariance assumption

The first method requires a certain size of validation data to accurately estimate $\mathbf{a}^*$. The estimation becomes difficult when the number of classes k is large. If we assume the equal covariance $\forall m \in [k], C_m = C$, the optimal $\mathbf{a}^*$ is calculated as $\mathbf{a}_i^* = \theta_i \left(1 - \frac{w_i}{\sum_{m=1}^k w_m} \right)$ where $w_i := \theta_i^2/n_i$.

Algorithm 1 Iterative Optimization SSL

Require: $\mathcal{X}_L = \cup_{i=1}^k \mathcal{X}_i$, $\mathcal{X}_U$, $\mathcal{X}_{val}$, update interval m , learning rate η , epochs E .

- 1: Initialize classifier $\hat{g}$ and risk parameter $\mathbf{a}$.
- 2: **for** $e = 1$ **to** E **do**
- 3: **if** $e \pmod{m} == 0$ **then**
- 4: Estimate covariance C_i using outputs of $\hat{g}$ on $\mathcal{X}_{val}$.
- 5: Update $\mathbf{a} \leftarrow \mathbf{a}^*$ minimizing variance (Theorem 3) on $\mathcal{X}_{val}$.
- 6: **end if**
- 7: Update $\hat{g} \leftarrow \hat{g} - \eta \nabla_{\hat{g}} \hat{R}_{lin}^a(\hat{g})$ using $\mathcal{X}_L \cup \mathcal{X}_U$.
- 8: **end for**
- 9: **return** $\hat{g}$

We use the linear risk with this $\mathbf{a}^*$ to learn $\hat{g}$. This does not require validation data to estimate $\mathbf{a}^*$.

Non-negative risk correction for complex models Note that we can express the linear risk estimator as

$$\hat{R}_{lin}^a(g) = \sum_{j=1}^k \left(a_j \hat{R}_{jj}(g) + \left(1 - \frac{a_j}{\theta_j} \right) \Delta_j(g) \right),$$

where $\Delta_j(g) := \hat{R}_{Uj}(g) - \sum_{i \neq j}^k \theta_i \hat{R}_{ij}(g)$. Here, $\Delta_j(g)$ can be viewed as a risk-rewriting estimator for $\theta_j \hat{R}_{jj}(g)$. As seen in PU and other risk-rewriting methods [Kiryo et al., 2017, Lu et al., 2020, Tang et al., 2023], when $\mathcal{G}$ consists of complex models such as deep neural networks, $\Delta_j(g)$ can become negative even when $\hat{R}_{jj}(g)$ is non-negative, which leads to overfitting. To mitigate this issue, similarly to the approach by Kiryo et al. [2017], we replace $\Delta_j(g)$ within $\hat{R}_{lin}^a(g)$ with a non-negative counterpart, $\Delta_j^{nn}(g) := \max\{0, \Delta_j(g)\}$, when training complex models.

6 EXPERIMENTS

6.1 VARIANCE COMPARISON OF R_{PN} , R_{PNU} AND R_{lin}^a

In this subsection, we empirically validate the variance reduction achieved by the proposed linear risk estimator $\hat{R}_{lin}^a(g)$. We compare the variance of three unbiased risk estimators for a pre-trained classifier g : the standard supervised risk estimator $\hat{R}(g)$ (PN risk, which corresponds to $\hat{R}_{lin}^a(g)$ where $\mathbf{a}_i = \theta_i$), the PNU risk estimator, and our proposed linear risk estimator with optimal parameter $\mathbf{a}^*$. We present binary classification results ($k = 2$) here; multi-class results are deferred to the Appendix.

Datasets. We use two datasets: a synthetic two-dimensional Gaussian dataset and Credit default dataset from the UCI Machine Learning Repository [Kelly et al.,

2023]. For the Gaussian dataset, the class-conditional distributions are $p_1(x) = \mathcal{N}((1, 1)^\top, I_2)$ and $p_2(x) = \mathcal{N}((0, 0)^\top, I_2)$.

Model and training. We train a logistic regression model $g(x) = w^\top x + b$ by minimizing the binary cross-entropy (BCE) loss with $n_1 = n_2 = 30$ labeled samples via stochastic gradient descent.

Evaluation loss. For the variance comparison, we use two loss functions: the BCE loss $l(g(x), y) = -y \log \sigma(g(x)) - (1 - y) \log(1 - \sigma(g(x)))$ as an asymmetric loss, and the 0-1 loss as a symmetric loss.

Evaluation protocol. With the trained classifier g fixed, we evaluate the variance of each risk estimator by repeatedly sampling from held-out data. To compute the optimal parameters η^* and a^* , we estimate the covariance matrices C_m for all $m \in [k]$ using the entire held-out data. For the PNU risk, we consider the set $S_{PNU} := S_{PNPU} \cup S_{PNNU}$ as defined in Theorem 9.

For each experimental condition (class prior θ_1 , unlabeled sample size n_U), we conduct 5,000 independent trials to estimate the risk variances. We vary $\theta_1 \in \{0.3, 0.5, 0.7\}$ and $n_U \in \{50, 100, 200, 500, 1000\}$. The labeled sample sizes for risk estimation are fixed at $n_1 = n_2 = 30$ across all settings.

6.1.1 Results

Figure 1 visualizes the variance ratios $\text{Var}(\hat{R}_{PNU})/\text{Var}(\hat{R})$ and $\text{Var}(\hat{R}_{lin}^{a^*})/\text{Var}(\hat{R})$ as a function of n_U for each class prior.

Effect of unlabeled sample size. As the number of unlabeled samples n_U increases, both the PNU and the proposed linear risk estimator achieve progressively lower variance ratios, confirming that unlabeled data effectively reduces the variance of risk estimation.

Effect of class prior. When the class prior is imbalanced ($\theta_1 = 0.3$ or $\theta_1 = 0.7$), both methods achieve substantial variance reduction. These results are consistent with Theorem 4. In the balanced case ($\theta_1 = 0.5$) with the asymmetric loss (BCE), the PNU risk shows only marginal improvement (e.g., a ratio of 0.953 at $n_U = 1,000$), whereas the proposed linear risk estimator still achieves a meaningful reduction (a ratio of 0.883), as predicted by Theorem 5.

Effect of loss function symmetry. When the loss function is symmetric, the variance ratios of the proposed estimator and the PNU estimator are nearly identical. This is expected, since their optimal variances coincide under symmetric loss, as established in Theorem 9.

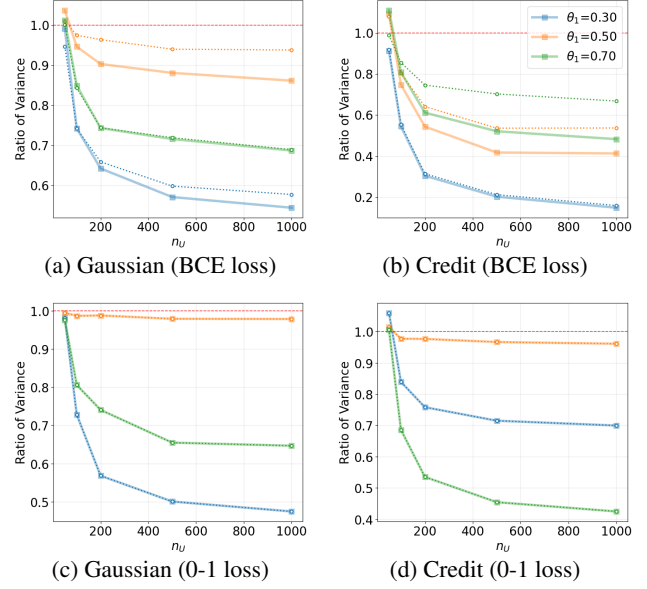


Figure 1: Variance ratio (each method / PN) as a function of the unlabeled sample size n_U for $\theta_1 \in \{0.3, 0.5, 0.7\}$. Solid lines denote the proposed linear risk estimator ($\hat{R}_{lin}^{a^*}$); dotted lines denote the PNU risk estimator. The red dashed line indicates the supervised baseline (ratio = 1). Both methods benefit from increasing n_U , and the proposed method consistently achieves equal or lower variance than PNU.

6.2 COMPARISON WITH EXISTING SEMI-SUPERVISED LEARNING METHODS

We compare the proposed SSL methods against four baselines on binary and multiclass classification tasks: (1) **Sup.**: standard supervised learning (linear risk with $a_i := \theta_i$); (2) **PNU**: the PNU risk estimator of Sakai et al. [2017] with the equal variance assumption (binary only); (3) **Pseudo Label (PL)**: self-training with pseudo-labels assigned by confidence thresholding, based on Yarowsky [1995]; and (4) **VAT**: virtual adversarial training [Miyato et al., 2019], which is a representative consistency regularization method. We denote our iterative method for optimizing g and a in this subsection as **Ours (Iter)**. For multiclass tasks, we additionally report **Ours (EC)**, the data-free method under the equal covariance assumption. We assume class priors θ_i are known. In practice, we can estimate the class priors using methods such as those proposed by Hideko KAWAKUBO [2016], Ramaswamy et al. [2016], Moreo et al. [2025], Lip-ton et al. [2018]. We use the cross-entropy loss for all methods. Detailed settings are provided in the Appendix.

Datasets. We use six binary and four multiclass classification tasks. *Binary tabular*: Adult, Banknote, Breast Cancer, and Credit Default from the UCI Machine Learning Repository [Kelly et al., 2023]. *Binary image*: MNIST

Table 1: Binary classification results (test accuracy %, mean $\pm$ std over 30 seeds). The best and second-best methods per row are **bolded** and underlined, respectively.

Dataset	(n_1, n_2)	Sup.	PNU	Ours (Iter)	PL	VAT
Adult	(15, 45)	<u>79.7$\pm$2.4</u>	79.1 $\pm$ 2.4	80.3$\pm$1.6	78.9 $\pm$ 1.8	78.4 $\pm$ 2.7
	(30, 30)	<u>80.4$\pm$2.1</u>	79.7 $\pm$ 2.5	80.8$\pm$1.5	78.2 $\pm$ 2.1	78.2 $\pm$ 2.0
	(50, 150)	<u>81.5$\pm$2.0</u>	80.5 $\pm$ 2.4	82.2$\pm$1.0	79.4 $\pm$ 1.9	79.6 $\pm$ 1.9
	(100, 100)	<u>81.5$\pm$1.8</u>	81.3 $\pm$ 2.1	82.4$\pm$1.0	78.9 $\pm$ 2.0	78.7 $\pm$ 2.2
Banknote	(15, 45)	95.6 $\pm$ 3.2	<u>96.9$\pm$2.0</u>	97.3$\pm$1.5	94.3 $\pm$ 4.4	95.7 $\pm$ 3.8
	(30, 30)	96.8 $\pm$ 2.4	<u>97.2$\pm$1.3</u>	97.8$\pm$1.1	95.7 $\pm$ 3.6	97.1 $\pm$ 1.6
	(50, 150)	<u>98.4$\pm$1.0</u>	<u>97.3$\pm$1.0</u>	98.6$\pm$0.9	97.2 $\pm$ 1.8	97.8 $\pm$ 1.0
	(100, 100)	<u>98.6$\pm$1.0</u>	98.0 $\pm$ 0.8	98.8$\pm$1.0	97.6 $\pm$ 1.1	98.0 $\pm$ 0.9
Breast Cancer	(15, 45)	93.6 $\pm$ 2.9	93.9 $\pm$ 2.8	94.5$\pm$2.3	93.4 $\pm$ 2.5	93.8 $\pm$ 2.9
	(30, 30)	<u>93.9$\pm$2.2</u>	<u>93.9$\pm$2.5</u>	94.4$\pm$1.8	93.1 $\pm$ 2.7	93.5 $\pm$ 3.0
	(50, 150)	<u>95.4$\pm$1.8</u>	94.9 $\pm$ 2.3	95.5$\pm$1.7	94.8 $\pm$ 2.1	95.1 $\pm$ 2.1
	(100, 100)	95.6$\pm$1.6	94.3 $\pm$ 2.6	95.6$\pm$1.8	<u>94.4$\pm$2.0</u>	94.3 $\pm$ 2.3
Credit	(15, 45)	79.0$\pm$1.1	<u>78.6$\pm$1.3</u>	78.2 $\pm$ 2.0	72.8 $\pm$ 4.6	71.7 $\pm$ 4.5
	(30, 30)	78.6 $\pm$ 1.6	79.2$\pm$1.0	<u>78.9$\pm$1.2</u>	66.3 $\pm$ 6.5	67.8 $\pm$ 5.6
	(50, 150)	79.5$\pm$1.2	79.5$\pm$0.9	<u>78.9$\pm$1.3</u>	72.1 $\pm$ 4.5	74.1 $\pm$ 3.9
	(100, 100)	79.8$\pm$1.1	<u>79.7$\pm$0.8</u>	79.3 $\pm$ 1.1	70.4 $\pm$ 5.3	72.5 $\pm$ 4.6
MNIST 4v9	(15, 45)	92.0 $\pm$ 3.9	<u>94.8$\pm$1.5</u>	94.7 $\pm$ 1.9	95.2$\pm$3.0	93.0 $\pm$ 3.3
	(30, 30)	94.1 $\pm$ 2.5	95.5 $\pm$ 1.0	<u>95.6$\pm$1.6</u>	96.5$\pm$1.7	95.2 $\pm$ 1.3
	(50, 150)	96.1 $\pm$ 1.3	96.9 $\pm$ 1.1	97.3$\pm$0.9	<u>97.2$\pm$0.8</u>	<u>97.2$\pm$0.6</u>
	(100, 100)	96.6 $\pm$ 1.7	97.3 $\pm$ 0.9	97.7$\pm$0.7	97.4 $\pm$ 1.4	<u>97.6$\pm$0.5</u>
CIFAR CvD	(15, 45)	54.3 $\pm$ 2.8	<u>55.0$\pm$2.4</u>	55.4$\pm$1.6	53.8 $\pm$ 2.5	54.6 $\pm$ 2.5
	(30, 30)	55.2 $\pm$ 2.6	56.0$\pm$2.1	55.8 $\pm$ 2.0	55.7 $\pm$ 2.6	55.4 $\pm$ 2.4
	(50, 150)	57.7 $\pm$ 2.5	58.3$\pm$2.1	<u>58.2$\pm$2.1</u>	57.1 $\pm$ 2.7	57.7 $\pm$ 2.2
	(100, 100)	59.2 $\pm$ 1.9	59.6$\pm$2.1	59.6$\pm$1.7	59.2 $\pm$ 1.5	<u>59.4$\pm$1.4</u>

digits 4 vs. 9 (MNIST 4v9) and CIFAR-10 cat vs. dog (CIFAR CvD). *Multiclass*: Covertypes, Shuttle, and Dry Beans (7 classes), and MNIST (10 classes). For binary tasks, labeled samples are specified as $(n_1, n_2) \in \{(15, 45), (30, 30), (50, 150), (100, 100)\}$ with $n_U = 300$ (tabular) or $n_U = 5,000$ (image). For multiclass tasks, we test three regimes of labeled sample sizes—*balanced* (equal samples per class), *mild imbalance*, and *severe imbalance*—with $n_U = 5,000$. Please refer to the Appendix for further details. The validation set consists of 60 samples for all settings, except for the multiclass image task, which uses 200 samples.

Models and training. We use multi-layer perceptrons (MLPs) for tabular datasets and convolutional neural networks (CNNs) for image datasets. All models are trained using the Adam optimizer. Since these models are complex, we apply non-negative risk correction in Section 5.2 to our methods and PNU learning.

6.2.1 Results

Binary classification results. Table 1 summarizes the binary classification results. Across tabular datasets, risk-rewriting methods (Ours and PNU) consistently outperform Pseudo Label and VAT, often by a large margin. This is expected, as tabular datasets typically do not satisfy the distributional assumptions relied upon by those methods. In almost all settings, our method ranks first or second.

Table 2: Multiclass classification results (test accuracy %, mean $\pm$ std over 30 seeds). The best and second-best methods per row are **bolded** and underlined, respectively.

Dataset	Regime (n)	Sup.	Ours (Iter)	Ours (EC)	PL	VAT
Covertypes	Bal. (70)	<u>52.5$\pm$3.1</u>	52.0 $\pm$ 2.9	53.1$\pm$3.4	42.1 $\pm$ 4.9	42.4 $\pm$ 4.6
	Bal. (140)	56.5$\pm$2.5	55.0 $\pm$ 2.6	<u>56.1$\pm$2.7</u>	46.7 $\pm$ 3.9	46.6 $\pm$ 3.1
	Bal. (350)	61.6$\pm$1.8	<u>61.2$\pm$2.1</u>	<u>61.2$\pm$1.9</u>	52.9 $\pm$ 2.4	51.8 $\pm$ 2.5
	Mild (140)	56.9 $\pm$ 2.8	<u>57.4$\pm$2.1</u>	57.7$\pm$1.8	49.6 $\pm$ 2.9	49.8 $\pm$ 2.2
	Mild (350)	62.0 $\pm$ 2.0	62.7$\pm$1.4	<u>62.3$\pm$1.7</u>	54.3 $\pm$ 2.3	54.7 $\pm$ 2.1
	Sev. (140)	58.8$\pm$2.3	<u>58.6$\pm$2.5</u>	58.8$\pm$2.0	53.7 $\pm$ 2.8	54.4 $\pm$ 2.3
Shuttle	Sev. (350)	63.2 $\pm$ 1.6	<u>63.3$\pm$1.7</u>	63.7$\pm$1.5	58.4 $\pm$ 2.5	58.2 $\pm$ 2.2
	Bal. (70)	94.3 $\pm$ 3.1	96.7$\pm$1.2	<u>96.2$\pm$1.3</u>	89.9 $\pm$ 4.6	95.5 $\pm$ 2.4
	Bal. (140)	94.9 $\pm$ 3.0	97.3$\pm$1.0	<u>97.1$\pm$1.0</u>	93.8 $\pm$ 3.8	96.7 $\pm$ 0.7
	Bal. (350)	97.2 $\pm$ 1.1	97.3 $\pm$ 1.0	97.4$\pm$0.7	96.9 $\pm$ 1.3	96.7 $\pm$ 1.2
	Mild (140)	96.9$\pm$1.1	96.9$\pm$1.0	96.9$\pm$0.9	96.2 $\pm$ 2.0	<u>96.5$\pm$1.3</u>
	Mild (350)	97.2 $\pm$ 1.0	97.6$\pm$1.1	<u>97.3$\pm$0.7</u>	97.0 $\pm$ 1.0	96.7 $\pm$ 0.7
DryBeans	Sev. (140)	97.0 $\pm$ 1.2	97.2$\pm$0.9	<u>97.1$\pm$1.7</u>	96.1 $\pm$ 1.6	96.0 $\pm$ 1.6
	Sev. (350)	97.3 $\pm$ 0.8	97.6$\pm$0.8	<u>97.4$\pm$1.0</u>	97.3 $\pm$ 0.8	96.8 $\pm$ 1.0
	Bal. (70)	86.4 $\pm$ 1.7	87.6 $\pm$ 1.6	86.6 $\pm$ 1.6	<u>88.1$\pm$2.0</u>	88.5$\pm$1.3
	Bal. (140)	88.8 $\pm$ 1.6	89.9 $\pm$ 0.9	89.2 $\pm$ 1.0	90.0$\pm$1.1	89.9 $\pm$ 1.4
	Bal. (350)	90.5 $\pm$ 0.7	91.3$\pm$0.6	90.4 $\pm$ 0.9	<u>90.8$\pm$0.9</u>	90.8 $\pm$ 1.0
	Mild (140)	88.8 $\pm$ 1.6	<u>89.1$\pm$1.7</u>	88.9 $\pm$ 1.6	88.7 $\pm$ 1.8	89.5$\pm$1.5
MNIST	Mild (350)	90.4 $\pm$ 1.2	<u>90.7$\pm$0.9</u>	<u>90.7$\pm$0.7</u>	90.6 $\pm$ 1.0	90.8$\pm$0.8
	Sev. (140)	84.2 $\pm$ 2.5	84.9 $\pm$ 2.6	<u>85.1$\pm$2.2</u>	83.4 $\pm$ 2.9	85.3$\pm$2.2
	Sev. (350)	88.1 $\pm$ 1.7	<u>89.1$\pm$1.5</u>	88.7 $\pm$ 1.4	88.4 $\pm$ 2.0	89.2$\pm$1.3
	Bal. (200)	86.9 $\pm$ 1.0	86.8 $\pm$ 1.7	<u>87.3$\pm$1.3</u>	89.7$\pm$1.6	87.1 $\pm$ 1.4
	Bal. (500)	91.9 $\pm$ 1.0	91.8 $\pm$ 1.4	92.2 $\pm$ 0.8	92.9$\pm$0.9	92.3 $\pm$ 0.7
	Bal. (1500)	95.7 $\pm$ 0.6	95.4 $\pm$ 0.6	95.4 $\pm$ 0.6	95.9$\pm$0.6	95.7 $\pm$ 0.5
MNIST	Mild (500)	89.9 $\pm$ 1.0	89.8 $\pm$ 1.4	<u>90.4$\pm$1.2</u>	91.7$\pm$1.5	90.1 $\pm$ 1.3
	Mild (1500)	94.8 $\pm$ 0.6	94.3 $\pm$ 0.8	94.4 $\pm$ 0.7	95.1$\pm$0.6	94.8 $\pm$ 0.6
	Sev. (500)	77.3 $\pm$ 2.5	81.7$\pm$3.0	<u>80.6$\pm$2.4</u>	79.6 $\pm$ 2.8	77.1 $\pm$ 2.6
	Sev. (1500)	88.5 $\pm$ 1.5	<u>89.5$\pm$1.7</u>	89.2 $\pm$ 1.4	90.7$\pm$2.2	88.4 $\pm$ 1.7

On MNIST 4v9, Pseudo Label benefits from the well-separated cluster structure of the MNIST digits, achieving the best accuracy with very few labels ($n_1=15$). However, on CIFAR-10 cat vs. dog, a more challenging task, it fails to yield comparable performance.

Multiclass classification results. Table 2 presents the multiclass results for the tabular datasets and MNIST. On Covertypes and Shuttle, our methods **Ours (Iter)** and **Ours (EC)** consistently rank first or second, while PL and VAT sometimes underperform compared to standard supervised learning.

For Dry Beans and MNIST, although PL and VAT show strong performance, our methods achieve better or comparable accuracy to standard supervised learning. In summary, our methods are more robust than PL and VAT, which rely on distributional assumptions.

It is worth highlighting that the data-free **Ours (EC)** method demonstrates highly competitive accuracy, matching or closely following the iterative optimization approach without requiring a validation set for parameter tuning.

7 CONCLUSIONS

We proposed a generalized framework for distribution-free semi-supervised learning based on risk rewriting. By formulating the set of linear combinations of component risks

S_{lin} , our framework subsumes existing methods such as PNU learning and naturally extends to multiclass classification. Experiments confirmed that our methods consistently match or outperform existing approaches.

PNU learning has had a major impact on subsequent research in SSL and weakly supervised learning [Ishida et al., 2017, Shimada et al., 2021, Hien et al., 2024, Sakai et al., 2018, Hammoudeh and Lowd, 2020, Wang et al., 2025, Kato and Yasui, 2020, Tsuchiya et al., 2021, Hayashi et al., 2018]. These studies attempt to reduce the variance of risks by simply mixing rewritten risks similarly to PNU learning. By applying our framework, these methods can be formulated using the linear combination set to construct risk estimators with smaller variances, thereby improving learning efficiency. We leave the application of our generalized framework to these broad problem settings for future work.

Acknowledgements

This work was supported in part by JSPS KAKENHI Grant Numbers JP24K14849 and JP26H02491.

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Appendix

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A OMITTED PROOFS

Proof of Theorem 1. We can write $R_{lin}^{\{a_{ij}\}, \{b_j\}}(g)$ as

$$\begin{aligned} R_{lin}^{\{a_{ij}\}, \{b_j\}}(g) &= \sum_{i,j} a_{ij} R_{ij}(g) + \sum_j b_j R_{Uj}(g) \\ &= \sum_{i,j} a_{ij} R_{ij}(g) + \sum_j b_j \left(\sum_i \theta_i R_{ij}(g) \right) \\ &= \sum_{i,j} a_{ij} R_{ij}(g) + \sum_{i,j} \theta_i b_j R_{ij}(g) \\ &= \sum_{i,j} (a_{ij} + \theta_i b_j) R_{ij}(g). \end{aligned}$$

In addition,

$$R(g) = \sum_{i=1}^k \theta_i R_{ii}(g) = \sum_{i,j} \delta_{ij} \theta_i R_{ij}(g).$$

Considering $\forall g \in \mathcal{G}$, $R_{lin}(g) = R(g)$,

$$\sum_{i,j} (a_{ij} + \theta_i b_j - \delta_{ij} \theta_i) R_{ij}(g) = 0 \quad (\forall g \in \mathcal{G}).$$

Then, under Assumption 1, $\forall i, \theta_i = a_{ii} + \theta_i b_i$ and $\forall i \neq j, a_{ij} + \theta_i b_j = 0$. Substituting $b_i = 1 - \frac{a_{ii}}{\theta_i}$ and $a_{ij} = -\theta_i b_j = \frac{\theta_i}{\theta_j} a_{jj} - \theta_i$ into $R_{lin}^{\{a_{ij}\}, \{b_j\}}(g)$, we derive the objective set. $\square$

Proof of Theorem 3. The minimum value of the convex quadratic function $f(\mathbf{a})$ is given by the algebraic identity:

$$f(\mathbf{a}^*) = c - \mathbf{b}^T A^{-1} \mathbf{b} \tag{3}$$

Step 1: Simplifying the Term $\mathbf{b}^T A^{-1} \mathbf{b}$ First, we express A and $\mathbf{b}$ in terms of S and Q .

$$\begin{aligned} A &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} Q C_m Q = Q \left(\sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m \right) Q = Q S Q \\ \mathbf{b} &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} Q C_m \mathbf{d}_m = Q \sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m \mathbf{d}_m \end{aligned}$$

Let $\mathbf{v} := \sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m \mathbf{d}_m$. Then $\mathbf{b} = Q\mathbf{v}$. Substituting these into the quadratic term in Eq. (3):

$$\begin{aligned} \mathbf{b}^T A^{-1} \mathbf{b} &= (Q\mathbf{v})^T (QSQ)^{-1} (Q\mathbf{v}) \\ &= \mathbf{v}^T Q (Q^{-1} S^{-1} Q^{-1}) Q \mathbf{v} \\ &= \mathbf{v}^T S^{-1} \mathbf{v} \end{aligned}$$

Next, we simplify $\mathbf{v}$ using $\mathbf{d}_m = \mathbf{1} - \mathbf{e}_m$:

$$\begin{aligned} \mathbf{v} &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m (\mathbf{1} - \mathbf{e}_m) \\ &= \left(\sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m \right) \mathbf{1} - \sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m \mathbf{e}_m \\ &= S\mathbf{1} - \mathbf{u} \end{aligned}$$

Thus, the reduction term is $\mathbf{v}^T S^{-1} \mathbf{v} = (S\mathbf{1} - \mathbf{u})^T S^{-1} (S\mathbf{1} - \mathbf{u})$.

Step 2: Expansion of $\mathbf{v}^T S^{-1} \mathbf{v}$ Expanding the term derived above:

$$\begin{aligned} \mathbf{v}^T S^{-1} \mathbf{v} &= (\mathbf{1}^T S - \mathbf{u}^T) S^{-1} (S\mathbf{1} - \mathbf{u}) \\ &= \mathbf{1}^T S S^{-1} S\mathbf{1} - \mathbf{u}^T S^{-1} S\mathbf{1} - \mathbf{1}^T S S^{-1} \mathbf{u} + \mathbf{u}^T S^{-1} \mathbf{u} \\ &= \mathbf{1}^T S\mathbf{1} - 2\mathbf{1}^T \mathbf{u} + \mathbf{u}^T S^{-1} \mathbf{u}. \end{aligned}$$

Step 3: Expansion of the Constant c We expand $c = \sum_{m=1}^k \frac{\theta_m^2}{n_m} \mathbf{d}_m^T C_m \mathbf{d}_m$:

$$\begin{aligned} c &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} (\mathbf{1} - \mathbf{e}_m)^T C_m (\mathbf{1} - \mathbf{e}_m) \\ &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} (\mathbf{1}^T C_m \mathbf{1} - 2\mathbf{1}^T C_m \mathbf{e}_m + \mathbf{e}_m^T C_m \mathbf{e}_m) \\ &= \mathbf{1}^T \left(\sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m \right) \mathbf{1} - 2\mathbf{1}^T \left(\sum_{m=1}^k \frac{\theta_m^2}{n_m} C_m \mathbf{e}_m \right) + \sum_{m=1}^k \frac{\theta_m^2}{n_m} \mathbf{e}_m^T C_m \mathbf{e}_m \\ &= \mathbf{1}^T S\mathbf{1} - 2\mathbf{1}^T \mathbf{u} + \sum_{m=1}^k \frac{\theta_m^2}{n_m} (C_m)_{mm} \end{aligned}$$

Final Calculation Substituting the expanded forms of c and $\mathbf{v}^T S^{-1} \mathbf{v}$ back into Eq. (3):

$$\begin{aligned} f(\mathbf{a}^*) &= c - \mathbf{v}^T S^{-1} \mathbf{v} \\ &= \left(\mathbf{1}^T S\mathbf{1} - 2\mathbf{1}^T \mathbf{u} + \sum_{m=1}^k \frac{\theta_m^2}{n_m} (C_m)_{mm} \right) - (\mathbf{1}^T S\mathbf{1} - 2\mathbf{1}^T \mathbf{u} + \mathbf{u}^T S^{-1} \mathbf{u}) \\ &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} (C_m)_{mm} - \mathbf{u}^T S^{-1} \mathbf{u} \end{aligned}$$

□

Proof of Theorem 4. Let us define the weight for class m as w_m and the total weight W :

$$w_m := \frac{\theta_m^2}{n_m}, \quad W := \sum_{m=1}^k w_m = \sum_{m=1}^k \frac{\theta_m^2}{n_m}.$$

We also define a weight vector $\mathbf{w} \in \mathbb{R}^k$ as $\mathbf{w} = [w_1, w_2, \dots, w_k]^T$. Under the assumption that $C_m = C$ for all m , the total weighted covariance matrix S simplifies to:

$$S = \sum_{m=1}^k w_m C_m = \left(\sum_{m=1}^k w_m \right) C = WC.$$

Similarly, the weighted covariance vector $\mathbf{u}$ simplifies to

$$\mathbf{u} = \sum_{m=1}^k w_m C_m \mathbf{e}_m = C \sum_{m=1}^k w_m \mathbf{e}_m = C\mathbf{w}.$$

Then we can write

$$\begin{aligned} \mathbf{u}^T S^{-1} \mathbf{u} &= (C\mathbf{w})^T (WC)^{-1} (C\mathbf{w}) = \mathbf{w}^T C \frac{1}{W} C^{-1} C\mathbf{w} \\ &= \frac{1}{W} \mathbf{w}^T C\mathbf{w}. \end{aligned}$$

Under the assumption that C has diagonal elements ρ_1 and off-diagonal elements ρ_2 , we have

$$C = (\rho_1 - \rho_2)I + \rho_2 \mathbf{1}\mathbf{1}^T.$$

Evaluating the quadratic form $\mathbf{w}^T C\mathbf{w}$ yields

$$\begin{aligned} \mathbf{w}^T C\mathbf{w} &= \mathbf{w}^T ((\rho_1 - \rho_2)I + \rho_2 \mathbf{1}\mathbf{1}^T) \mathbf{w} = (\rho_1 - \rho_2)|\mathbf{w}|^2 + \rho_2(\mathbf{w}^T \mathbf{1})^2 \\ &= (\rho_1 - \rho_2) \sum_{m=1}^k w_m^2 + \rho_2 W^2, \end{aligned}$$

This completes the proof of the equality. To derive the upper bound, we note that $\sum_{m=1}^k w_m^2 \leq (\max_m w_m) \sum_{m=1}^k w_m = W \max_m w_m$. Since $\rho_1 > \rho_2$, we have

$$\begin{aligned} \mathbf{u}^T S^{-1} \mathbf{u} &\leq (\rho_1 - \rho_2) \frac{W \max_m w_m}{W} + \rho_2 W \\ &= \rho_1 \max_m w_m - \rho_2 \max_m w_m + \rho_2 W \\ &= \rho_1 \max_{m \in [k]} w_m + \rho_2 \left(W - \max_{m \in [k]} w_m \right). \end{aligned}$$

□

Proof of Theorem 5. Dominance inequality (1): Recall that

$$\begin{aligned} R_{\text{PNPU}}^\eta(g) &= (1 - \eta)(\theta_1 R_{11} + \theta_2 R_{22}) + \eta(\theta_1 R_{11} - \theta_1 R_{12} + R_{U2}) \\ &= \theta_1 R_{11} + (1 - \eta)\theta_2 R_{22} - \eta\theta_1 R_{12} + \eta R_{U2}. \end{aligned}$$

This risk is equivalent to $R_{lin}^a(g)$ where $(a_1, a_2) = (\theta_1, \theta_2(1 - \eta))$. Similarly,

$$\begin{aligned} R_{\text{PNNU}}^\eta(g) &= (1 - \eta)(\theta_1 R_{11} + \theta_2 R_{22}) + \eta(R_{U2} - \theta_2 R_{21} + \theta_2 R_{22}) \\ &= (1 - \eta)\theta_1 R_{11} + \theta_2 R_{22} - \eta\theta_2 R_{21} + \eta R_{U2}, \end{aligned}$$

which is equivalent to $R_{lin}^a(g)$ where $(a_1, a_2) = (\theta_1(1 - \eta), \theta_2)$.

The sets of estimators for PNPU and PNNU learning are subsets of the generalized linear estimator set S_{lin} . Thus, Eq. (1) holds.

Derivation of PNPU Minimum Variance: We explicitly derive the minimum variance for the PNPU estimator in the binary setting ($k = 2$). The variance of the estimator $\hat{R}_{\text{PNPU}}^\eta$ when $n_U \rightarrow \infty$ is given by:

$$\begin{aligned} \text{Var}(\hat{R}_{\text{PNPU}}^\eta) &= \text{Var}(\theta_1 \hat{R}_{11} - \eta\theta_1 \hat{R}_{12} + (1 - \eta)\theta_2 \hat{R}_{22}) \\ &= \theta_1^2 \text{Var}(\hat{R}_{11} - \eta\hat{R}_{12}) + (1 - \eta)^2 \theta_2^2 \text{Var}(\hat{R}_{22}). \end{aligned}$$

Assuming $C_1 = C_2 = C$ with diagonal elements ρ_1 and off-diagonal elements ρ_2 :

$$\begin{aligned}\text{Var}(\hat{R}_{11} - \eta \hat{R}_{12}) &= \text{Var}(\hat{R}_{11}) - 2\eta \text{Cov}(\hat{R}_{11}, \hat{R}_{12}) + \eta^2 \text{Var}(\hat{R}_{12}) \\ &= \frac{\rho_1}{n_1} - \frac{2\eta\rho_2}{n_1} + \frac{\eta^2\rho_1}{n_1}.\end{aligned}$$

Substituting this back into the total variance expression and using $w_i = \theta_i^2/n_i$:

$$\begin{aligned}\text{Var}(\hat{R}_{\text{PNPU}}^\eta) &= \theta_1^2 \left(\frac{\rho_1 - 2\eta\rho_2 + \eta^2\rho_1}{n_1} \right) + (1 - \eta)^2 \theta_2^2 \frac{\rho_1}{n_2} \\ &= w_1(\rho_1 - 2\eta\rho_2 + \eta^2\rho_1) + w_2(1 - 2\eta + \eta^2)\rho_1 \\ &= (w_1 + w_2)\rho_1\eta^2 - 2(w_1\rho_2 + w_2\rho_1)\eta + (w_1 + w_2)\rho_1.\end{aligned}$$

Then,

$$\begin{aligned}\text{Var}(\hat{R}_{\text{PNPU}}^{\eta*}) &= (w_1 + w_2)\rho_1 - \frac{(-2(w_1\rho_2 + w_2\rho_1))^2}{4(w_1 + w_2)\rho_1} \\ &= (w_1 + w_2)\rho_1 - \frac{4(w_1\rho_2 + w_2\rho_1)^2}{4(w_1 + w_2)\rho_1} \\ &= \frac{(w_1 + w_2)^2\rho_1^2 - (w_1\rho_2 + w_2\rho_1)^2}{\rho_1(w_1 + w_2)} \\ &= \frac{w_1^2(\rho_1^2 - \rho_2^2) + 2w_1w_2\rho_1(\rho_1 - \rho_2)}{\rho_1(w_1 + w_2)}.\end{aligned}$$

Variance gap (2): Using the result from Theorem 4 for $k = 2$, the minimum variance of our proposed estimator is:

$$\begin{aligned}\text{Var}(\hat{R}_{lin}^{a*}) &= \sum_{m=1}^2 w_m\rho_1 - \left[(\rho_1 - \rho_2) \frac{w_1^2 + w_2^2}{W} + \rho_2 W \right] \\ &= \rho_1 W - \rho_2 W - (\rho_1 - \rho_2) \frac{w_1^2 + w_2^2}{W} \\ &= (\rho_1 - \rho_2) \left(W - \frac{w_1^2 + w_2^2}{W} \right) \\ &= (\rho_1 - \rho_2) \frac{(w_1 + w_2)^2 - (w_1^2 + w_2^2)}{W} \\ &= (\rho_1 - \rho_2) \frac{2w_1w_2}{w_1 + w_2}.\end{aligned}$$

We compute the gap $\text{Var}(\hat{R}_{\text{PNPU}}^{\eta*}) - \text{Var}(\hat{R}_{lin}^{a*})$:

$$\begin{aligned}\text{Gap} &= \frac{w_1^2(\rho_1 - \rho_2)(\rho_1 + \rho_2) + 2w_1w_2\rho_1(\rho_1 - \rho_2)}{\rho_1 W} - \frac{2w_1w_2(\rho_1 - \rho_2)}{W} \\ &= \frac{(\rho_1 - \rho_2)}{W} \left[\frac{w_1^2(\rho_1 + \rho_2) + 2w_1w_2\rho_1}{\rho_1} - 2w_1w_2 \right] \\ &= \frac{(\rho_1 - \rho_2)}{W} \left[\frac{w_1^2(\rho_1 + \rho_2)}{\rho_1} + 2w_1w_2 - 2w_1w_2 \right] \\ &= \frac{w_1^2(\rho_1^2 - \rho_2^2)}{(w_1 + w_2)\rho_1}.\end{aligned}$$

By symmetry, the gap for PNNU is $\frac{w_2^2(\rho_1^2 - \rho_2^2)}{(w_1 + w_2)\rho_1}$. The result follows. $\square$

Proof of Theorem 6. From the definition of symmetric loss $\sum_{j=1}^k R_{ij} = \alpha$, the following holds:

$$R_{ik} = \alpha - \sum_{j=1}^{k-1} R_{ij}, \quad R_{Uk} = \alpha - \sum_{j=1}^{k-1} R_{Uj}$$

Substituting the above expressions into $R_{lin}(g)$, we derive

$$\begin{aligned}
R_{lin}^{\{a_{ij}\}, \{b_j\}} &= \sum_{i,j} a_{ij} R_{ij}(g) + \sum_j b_j R_{Uj} \\
&= \sum_i \left[\sum_{j=1}^{k-1} a_{ij} R_{ij} + a_{ik} \left(\alpha - \sum_{j=1}^{k-1} R_{ij} \right) \right] + \sum_{j=1}^{k-1} b_j R_{Uj} + b_k \left(\alpha - \sum_{j=1}^{k-1} R_{Uj} \right) \\
&= \sum_i \sum_{j=1}^{k-1} \underbrace{(a_{ij} - a_{ik})}_{\tilde{a}_{ij}} R_{ij} + \sum_{j=1}^{k-1} \underbrace{(b_j - b_k)}_{\tilde{b}_j} R_{Uj} + \alpha \left(\sum_i a_{ik} + b_k \right)
\end{aligned}$$

where we define $\tilde{a}_{ij} := a_{ij} - a_{ik}$ and $\tilde{b}_j := b_j - b_k$. Using $R_{Uj} = \sum_i \theta_i R_{ij}$, the second term can be rewritten as:

$$\sum_{j=1}^{k-1} (b_j - b_k) R_{Uj} = \sum_{j=1}^{k-1} (b_j - b_k) \sum_i \theta_i R_{ij} = \sum_i \sum_{j=1}^{k-1} (b_j - b_k) \theta_i R_{ij}$$

Substituting these back into R_{lin} and rearranging:

$$R_{lin}^{\{a_{ij}\}, \{b_j\}} = \sum_i \sum_{j=1}^{k-1} (a_{ij} - a_{ik} + (b_j - b_k) \theta_i) R_{ij} + \alpha \left(\sum_i a_{ik} + b_k \right)$$

On the other hand, the true risk $R(g)$ can be written as follows (expanding R_{ik}):

$$\begin{aligned}
R(g) &= \sum_i \theta_i R_{ii} = \sum_{i=1}^{k-1} \theta_i R_{ii} + \theta_k R_{kk} \\
&= \sum_{i=1}^{k-1} \theta_i R_{ii} + \theta_k \left(\alpha - \sum_{j=1}^{k-1} R_{kj} \right)
\end{aligned}$$

Under Assumption 2, we derive the conditions for the identity $\forall g \in \mathcal{G}, R_{lin}(g) = R(g)$ to hold :

$$\begin{cases} \forall i, j \in [k-1], & a_{ij} - a_{ik} + (b_j - b_k) \theta_i - \delta_{ij} \theta_i = 0 \quad (\text{Coeff. of } R_{ij}) \\ \forall j \in [k-1], & a_{kj} - a_{kk} + (b_j - b_k) \theta_k + \theta_k = 0 \quad (\text{Coeff. of } R_{kj}) \\ & \sum_i a_{ik} + b_k - \theta_k = 0 \quad (\text{Constant term}) \end{cases}$$

Assume that arbitrary $\tilde{a}_{ij} \in \mathbb{R}$ and $\tilde{b}_j \in \mathbb{R}, \forall i \in [k], j \in [k-1]$ are given. There are values of a_{ij}, b_j that satisfy the above equations and $\tilde{a}_{ij} = a_{ij} - a_{ik}$ and $\tilde{b}_j = b_j - b_k$. Thus, we can simplify $R_{lin}^{\{a_{ij}\}, \{b_j\}}(g)$ as

$$R_{lin}(g) = \sum_i \sum_{j=1}^{k-1} \tilde{a}_{ij} R_{ij} + \sum_{j=1}^{k-1} \tilde{b}_j R_{Uj} + \alpha \theta_k, \quad \text{where } \tilde{a}_{ij}, \tilde{b}_j \in \mathbb{R}$$

The conditions are:

1. $\tilde{a}_{ij} + \tilde{b}_j \theta_i - \delta_{ij} \theta_i = 0 \quad \forall i, j \in [k-1]$
2. $\tilde{a}_{kj} + \tilde{b}_j \theta_k + \theta_k = 0 \quad \forall j \in [k-1]$

Solving these yields:

$$\begin{aligned}
\forall i \in [k-1], \quad & \tilde{a}_{ii} + \tilde{b}_i \theta_i - \theta_i = 0 \\
\Rightarrow \quad & \tilde{b}_i = 1 - \frac{\tilde{a}_{ii}}{\theta_i}
\end{aligned}$$

Also, when $i \neq j$, $\tilde{a}_{ij} + \tilde{b}_j \theta_i = 0$. From these, we get:

$$\begin{aligned} \forall i, j \in [k-1] \text{ s.t. } i \neq j, \quad \tilde{a}_{ij} &= -\theta_i \tilde{b}_j \\ &= -\theta_i \left(1 - \frac{\tilde{a}_{jj}}{\theta_j}\right) \end{aligned}$$

Furthermore, for the k -th class:

$$\begin{aligned} \tilde{a}_{kj} &= -\theta_k(1 + \tilde{b}_j) \\ &= -\theta_k \left(2 - \frac{\tilde{a}_{jj}}{\theta_j}\right) \end{aligned}$$

Substituting the above equations into $R_{lin}(g)$, we derive the target set. $\square$

Proof of Theorem 7. We derive the variance of the estimator $\hat{R}_{lin}^a$ under the symmetric loss assumption. Recall that the estimator is given by:

$$\hat{R}_{lin}^a = \sum_{i=1}^k \theta_i \hat{R}_{ii} + \sum_{i=1}^k \sum_{j=1}^{k-1} \theta_i \left(\frac{a_j}{\theta_j} - 1 \right) \hat{R}_{ij} - \sum_{j=1}^{k-1} \left(\frac{a_j}{\theta_j} - 1 \right) \hat{R}_{Uj}.$$

In the asymptotic regime where $n_U \rightarrow \infty$, the variance of the unlabeled component vanishes. Thus, we focus on the labeled components. Since the labeled datasets $\mathcal{X}_m$ for $m \in \{1, \dots, k\}$ are independent, the total variance is the sum of the variances contributed by each class. We define T_m as the terms in $\hat{R}_{lin}^a$ dependent on $\mathcal{X}_m$:

$$T_m = \theta_m \hat{R}_{mm} + \sum_{j=1}^{k-1} \theta_m \left(\frac{a_j}{\theta_j} - 1 \right) \hat{R}_{mj}.$$

Let $\check{\mathbf{R}}_m = [\hat{R}_{m1}, \dots, \hat{R}_{m,k-1}]^\top$ be the vector of component risks for class m restricted to the first $k-1$ labels. The covariance matrix of $\check{\mathbf{R}}_m$ is $\frac{1}{n_m} \check{C}_m$. We express T_m as a linear combination $\mathbf{c}_m^\top \check{\mathbf{R}}_m + \text{const}$ and determine the coefficient vector $\mathbf{c}_m \in \mathbb{R}^{k-1}$.

Case 1: $m < k$. For $m \in \{1, \dots, k-1\}$, the term $\hat{R}_{mm}$ is explicitly contained in the sum $\sum_{j=1}^{k-1}$. The coefficient for $\hat{R}_{mj}$ where $j \neq m$ is $\theta_m(a_j/\theta_j - 1)$. The coefficient for $\hat{R}_{mm}$ combines the first term and the summation term: $\theta_m + \theta_m(a_m/\theta_m - 1) = a_m$. Thus, the j -th element of $\mathbf{c}_m$ is:

$$(\mathbf{c}_m)_j = \begin{cases} \theta_m \left(\frac{a_j}{\theta_j} - 1 \right) & j \neq m \\ \theta_m \left(\frac{a_m}{\theta_m} \right) & j = m \end{cases}.$$

Using the diagonal matrix $\check{Q} = \text{diag}(1/\theta_1, \dots, 1/\theta_{k-1})$ and vector $\check{\mathbf{d}}_m = \mathbf{1} - \mathbf{e}_m$, we can write this compactly as:

$$\mathbf{c}_m = \theta_m(\check{Q}\mathbf{a} - \check{\mathbf{d}}_m).$$

Case 2: $m = k$. For class k , $\hat{R}_{kk}$ is not included in $\check{\mathbf{R}}_k$. We invoke the symmetric loss property $\sum_{j=1}^k \hat{R}_{kj} = \alpha$ (constant) to substitute $\hat{R}_{kk} = \alpha - \sum_{j=1}^{k-1} \hat{R}_{kj}$. Substituting this into T_k :

$$\begin{aligned} T_k &= \theta_k \left(\alpha - \sum_{j=1}^{k-1} \hat{R}_{kj} \right) + \sum_{j=1}^{k-1} \theta_k \left(\frac{a_j}{\theta_j} - 1 \right) \hat{R}_{kj} \\ &= \text{const} + \sum_{j=1}^{k-1} \left[-\theta_k + \theta_k \left(\frac{a_j}{\theta_j} - 1 \right) \right] \hat{R}_{kj} \\ &= \text{const} + \sum_{j=1}^{k-1} \theta_k \left(\frac{a_j}{\theta_j} - 2 \right) \hat{R}_{kj}. \end{aligned}$$

Here, the coefficient vector corresponds to $\check{\mathbf{d}}_k = 2\mathbf{1}$. Thus:

$$\mathbf{c}_k = \theta_k(\check{Q}\mathbf{a} - 2\mathbf{1}) = \theta_k(\check{Q}\mathbf{a} - \check{\mathbf{d}}_k).$$

Total Variance. The variance of the estimator is the sum of variances of T_m :

$$\begin{aligned} \text{Var}[\hat{R}_{lin}^{\mathbf{a}}] &= \sum_{m=1}^k \text{Var}[T_m] = \sum_{m=1}^k \mathbf{c}_m^\top \left(\frac{1}{n_m} \check{C}_m \right) \mathbf{c}_m \\ &= \sum_{m=1}^k \frac{\theta_m^2}{n_m} (\check{Q}\mathbf{a} - \check{\mathbf{d}}_m)^\top \check{C}_m (\check{Q}\mathbf{a} - \check{\mathbf{d}}_m). \end{aligned}$$

□

Proof of Theorem 8. First, we simplify the reduction term $\check{\mathbf{b}}^T \check{A}^{-1} \check{\mathbf{b}}$. Substituting $\check{A}^{-1} = \check{Q}^{-1} \check{S}^{-1} \check{Q}^{-1}$ and $\check{\mathbf{b}} = \check{Q}(\check{S}\mathbf{1} - \check{\mathbf{u}})$:

$$\begin{aligned} \check{\mathbf{b}}^T \check{A}^{-1} \check{\mathbf{b}} &= (\check{S}\mathbf{1} - \check{\mathbf{u}})^T \check{S}^{-1} (\check{S}\mathbf{1} - \check{\mathbf{u}}) \\ &= (\mathbf{1}^T \check{S} - \check{\mathbf{u}}^T)(\mathbf{1} - \check{S}^{-1} \check{\mathbf{u}}) \\ &= \mathbf{1}^T \check{S} \mathbf{1} - 2\mathbf{1}^T \check{\mathbf{u}} + \check{\mathbf{u}}^T \check{S}^{-1} \check{\mathbf{u}}. \end{aligned}$$

Next, we expand the constant term $\check{c}$. Recalling $\check{\mathbf{d}}_m = \mathbf{1} - \mathbf{e}_m$ for $m < k$ and $\check{\mathbf{d}}_k = 2\mathbf{1}$:

$$\begin{aligned} \check{c} &= \sum_{m=1}^{k-1} \frac{\theta_m^2}{n_m} (\mathbf{1} - \mathbf{e}_m)^T \check{C}_m (\mathbf{1} - \mathbf{e}_m) + \frac{\theta_k^2}{n_k} (2\mathbf{1})^T \check{C}_k (2\mathbf{1}) \\ &= \sum_{m=1}^{k-1} \frac{\theta_m^2}{n_m} (\mathbf{1}^T \check{C}_m \mathbf{1} - 2\mathbf{1}^T \check{C}_m \mathbf{e}_m + (\check{C}_m)_{mm}) + 4 \frac{\theta_k^2}{n_k} \mathbf{1}^T \check{C}_k \mathbf{1}. \end{aligned}$$

We regroup the terms using $\check{S} = \sum_{m=1}^{k-1} \frac{\theta_m^2}{n_m} \check{C}_m + \frac{\theta_k^2}{n_k} \check{C}_k$:

- Quadratic terms in $\mathbf{1}$:

$$\mathbf{1}^T \left(\sum_{m=1}^{k-1} \frac{\theta_m^2}{n_m} \check{C}_m \right) \mathbf{1} + 4 \frac{\theta_k^2}{n_k} \mathbf{1}^T \check{C}_k \mathbf{1} = \mathbf{1}^T (\check{S} - \frac{\theta_k^2}{n_k} \check{C}_k) \mathbf{1} + 4 \frac{\theta_k^2}{n_k} \mathbf{1}^T \check{C}_k \mathbf{1} = \mathbf{1}^T \check{S} \mathbf{1} + 3 \frac{\theta_k^2}{n_k} \mathbf{1}^T \check{C}_k \mathbf{1}.$$

- Linear terms in $\mathbf{1}$:

$$-2\mathbf{1}^T \sum_{m=1}^{k-1} \frac{\theta_m^2}{n_m} \check{C}_m \mathbf{e}_m = -2\mathbf{1}^T (\check{\mathbf{u}} + \frac{\theta_k^2}{n_k} \check{C}_k \mathbf{1}) = -2\mathbf{1}^T \check{\mathbf{u}} - 2 \frac{\theta_k^2}{n_k} \mathbf{1}^T \check{C}_k \mathbf{1}.$$

- Remaining diagonal terms: $\sum_{m=1}^{k-1} \frac{\theta_m^2}{n_m} (\check{C}_m)_{mm}$.

Thus:

$$\begin{aligned} \check{c} &= \mathbf{1}^T \check{S} \mathbf{1} - 2\mathbf{1}^T \check{\mathbf{u}} + \left(\sum_{m=1}^{k-1} \frac{\theta_m^2}{n_m} (\check{C}_m)_{mm} + \frac{\theta_k^2}{n_k} \mathbf{1}^T \check{C}_k \mathbf{1} \right) \\ &= \mathbf{1}^T \check{S} \mathbf{1} - 2\mathbf{1}^T \check{\mathbf{u}} + \sum_{m=1}^k \frac{\theta_m^2}{n_m} (C_m)_{mm} \end{aligned}$$

Subtracting the reduction term from $\check{c}$ cancels the $\mathbf{1}^T \check{S} \mathbf{1}$ and linear terms, yielding the result. □

Proof of Theorem 9. First, we derive the explicit forms of the risks under the symmetric loss assumption. Recall the symmetric condition implies $R_{12} = \alpha - R_{11}$ and $R_{21} = \alpha - R_{22}$. Substituting these into the definitions of PU and NU risks, we obtain:

$$\begin{aligned} R_{\text{PU}}(g) &= \theta_1 R_{11}(g) - \theta_1(\alpha - R_{11}(g)) + R_{U2}(g) \\ &= 2\theta_1 R_{11}(g) + R_{U2}(g) - \theta_1 \alpha, \end{aligned} \quad (4)$$

$$\begin{aligned} R_{\text{NU}}(g) &= \theta_2 R_{22}(g) - \theta_2(\alpha - R_{22}(g)) + R_{U1}(g) \\ &= 2\theta_2 R_{22}(g) + R_{U1}(g) - \theta_2 \alpha. \end{aligned} \quad (5)$$

Next, we consider the PNPU estimator $R_{\text{PNPU}}^\eta = (1 - \eta)R_{\text{PN}} + \eta R_{\text{PU}}$. Substituting (4):

$$\begin{aligned} R_{\text{PNPU}}^\eta(g) &= (1 - \eta)(\theta_1 R_{11} + \theta_2 R_{22}) + \eta(2\theta_1 R_{11} + R_{U2} - \theta_1 \alpha) \\ &= \theta_1(1 + \eta)R_{11} + \theta_2(1 - \eta)R_{22} + \eta R_{U2} - \eta \theta_1 \alpha. \end{aligned} \quad (6)$$

Similarly, for the PNNU estimator $R_{\text{PNNU}}^\eta = (1 - \eta)R_{\text{PN}} + \eta R_{\text{NU}}$, using (5):

$$\begin{aligned} R_{\text{PNNU}}^\eta(g) &= (1 - \eta)(\theta_1 R_{11} + \theta_2 R_{22}) + \eta(2\theta_2 R_{22} + R_{U1} - \theta_2 \alpha) \\ &= \theta_1(1 - \eta)R_{11} + \theta_2(1 + \eta)R_{22} + \eta R_{U1} - \eta \theta_2 \alpha. \end{aligned} \quad (7)$$

Now, we examine the general linear estimator set S_{lin} for $k = 2$ with symmetric loss. From Theorem 6, the estimator is parametrized by a scalar a_1 :

$$R_{lin}^{a_1}(g) = a_1 R_{11} + \theta_2 R_{22} + \theta_2 \left(\frac{a_1}{\theta_1} - 1 \right) R_{21} - \left(\frac{a_1}{\theta_1} - 1 \right) R_{U1}.$$

Using the identity $R_{21} = \alpha - R_{22}$ and $R_{U1} + R_{U2} = \alpha$ to align terms with (6), we rewrite $R_{lin}^{a_1}$:

$$\begin{aligned} R_{lin}^{a_1}(g) &= a_1 R_{11} + \theta_2 R_{22} + \theta_2 \left(\frac{a_1}{\theta_1} - 1 \right) (\alpha - R_{22}) - \left(\frac{a_1}{\theta_1} - 1 \right) (\alpha - R_{U2}) \\ &= a_1 R_{11} + \left[\theta_2 - \theta_2 \left(\frac{a_1}{\theta_1} - 1 \right) \right] R_{22} + \left(\frac{a_1}{\theta_1} - 1 \right) R_{U2} + \text{const.} \\ &= a_1 R_{11} + \theta_2 \left(2 - \frac{a_1}{\theta_1} \right) R_{22} + \left(\frac{a_1}{\theta_1} - 1 \right) R_{U2} + \text{const.} \end{aligned}$$

Comparing this with the PNPU form in (6), we see that they match if we set $a_1 = \theta_1(1 + \eta)$. Specifically:

- Coefficient of R_{11} : $a_1 \iff \theta_1(1 + \eta)$
- Coefficient of R_{22} : $\theta_2(2 - (1 + \eta)) = \theta_2(1 - \eta)$
- Coefficient of R_{U2} : $(1 + \eta) - 1 = \eta$

Since this mapping is bijective for $\eta, a_1 \in \mathbb{R}$, we have $S_{lin} = S_{\text{PNPU}}$.

Similarly, comparing $R_{lin}^{a_1}$ with PNNU form in (7) by setting $a_1 = \theta_1(1 - \eta)$ shows that $S_{lin} = S_{\text{PNNU}}$. Thus, all three sets are equivalent. $\square$

Proof of Theorem 10. We consider a hypothesis class $\mathcal{G}$ and its minimal ν -cover C_ν with respect to the L_∞ norm. By definition, $|C_\nu| = \mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_\infty)$, and for any $g \in \mathcal{G}$, there exists a $g' \in C_\nu$ such that $\|g - g'\|_\infty \leq \nu$.

Let $\hat{g}$ be the empirical risk minimizer and g^* be the true risk minimizer. As $n_U \rightarrow \infty$, the variance of the unlabeled risk components vanishes, so we focus on the empirical estimates derived from the labeled datasets.

1. Decomposition of Excess Risk and Covering Bound First, we relate the excess risk to the maximum estimation error over $\mathcal{G}$:

$$\begin{aligned} R(\hat{g}) - R(g^*) &= R(\hat{g}) - \hat{R}_{lin}^a(\hat{g}) + \hat{R}_{lin}^a(\hat{g}) - \hat{R}_{lin}^a(g^*) + \hat{R}_{lin}^a(g^*) - R(g^*) \\ &\leq R(\hat{g}) - \hat{R}_{lin}^a(\hat{g}) + \hat{R}_{lin}^a(g^*) - R(g^*) \quad (\text{since } \hat{R}_{lin}^a(\hat{g}) \leq \hat{R}_{lin}^a(g^*)) \\ &\leq 2 \sup_{g \in \mathcal{G}} |\hat{R}_{lin}^a(g) - R(g)|. \end{aligned}$$

Using the ν -cover C_ν , we can bound the deviation for any $g \in \mathcal{G}$ by mapping it to its closest element $g' \in C_\nu$. Because the loss function is L -Lipschitz continuous, we have $|l(g(x), j) - l(g'(x), j)| \leq L\|g(x) - g'(x)\|_\infty \leq L\nu$.

For the empirical risk estimator, the difference is bounded by:

$$\begin{aligned} |\hat{R}_{lin}^{\mathbf{a}}(g) - \hat{R}_{lin}^{\mathbf{a}}(g')| &\leq \sum_{m=1}^k \frac{\theta_m}{n_m} \sum_{q=1}^{n_m} \left(|l(g(x_q^m), m) - l(g'(x_q^m), m)| + \sum_{j=1}^k \left| \frac{a_j}{\theta_j} - 1 \right| |l(g(x_q^m), j) - l(g'(x_q^m), j)| \right) \\ &\leq \sum_{m=1}^k \theta_m (L\nu + kc_{\mathbf{a}}L\nu) = L\nu(1 + kc_{\mathbf{a}}) = L_{\mathbf{a}}\nu. \end{aligned}$$

Similarly, for the true risk, $|R(g) - R(g')| \leq \mathbb{E}[|l(g(x), y) - l(g'(x), y)|] \leq L\nu \leq L_{\mathbf{a}}\nu$. Thus, using the triangle inequality for any fixed $g \in \mathcal{G}$ and its closest element $g' \in C_\nu$ (where $\|g - g'\|_\infty \leq \nu$), we have:

$$\begin{aligned} |\hat{R}_{lin}^{\mathbf{a}}(g) - R(g)| &\leq |\hat{R}_{lin}^{\mathbf{a}}(g) - \hat{R}_{lin}^{\mathbf{a}}(g')| + |\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| + |R(g') - R(g)| \\ &\leq 2L_{\mathbf{a}}\nu + |\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| \\ &\leq 2L_{\mathbf{a}}\nu + \max_{g'' \in C_\nu} |\hat{R}_{lin}^{\mathbf{a}}(g'') - R(g'')|. \end{aligned}$$

Since the right-hand side is independent of the choice of g , we can now take the supremum over all $g \in \mathcal{G}$ on the left-hand side:

$$\sup_{g \in \mathcal{G}} |\hat{R}_{lin}^{\mathbf{a}}(g) - R(g)| \leq \max_{g' \in C_\nu} |\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| + 2L_{\mathbf{a}}\nu. \quad (8)$$

2. Bernstein's Inequality on the Finite Cover We now bound the estimation error for the finite set of hypotheses in C_ν . For any fixed $g' \in C_\nu$, the estimator $\hat{R}_{lin}^{\mathbf{a}}(g')$ is a sum of independent random variables:

$$\hat{R}_{lin}^{\mathbf{a}}(g') = \sum_{m=1}^k \sum_{q=1}^{n_m} \frac{\theta_m}{n_m} \underbrace{\left(l(g'(x_q^m), m) + \sum_{j=1}^k \left(\frac{a_j}{\theta_j} - 1 \right) l(g'(x_q^m), j) \right)}_{Z_q^m(g')} + \text{const.}$$

Since $0 \leq l(\cdot, \cdot) \leq c_l$, the variables $Z_q^m(g')$ are bounded by $\max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) kc_l(1 + c_{\mathbf{a}}) = \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) B_{\mathbf{a}}$.

First, for any fixed $g' \in C_\nu$, applying Bernstein's inequality yields:

$$P\left(|\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| > \epsilon\right) \leq 2 \exp\left(\frac{-\epsilon^2}{2 \text{Var}[\hat{R}_{lin}^{\mathbf{a}}(g')] + \frac{2}{3} \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) B_{\mathbf{a}} \epsilon}\right).$$

Since $\text{Var}[\hat{R}_{lin}^{\mathbf{a}}(g')] \leq \sigma_{\max}^2(\mathbf{a}, \nu)$ for all $g' \in C_\nu$, we can upper bound this probability by:

$$P\left(|\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| > \epsilon\right) \leq 2 \exp\left(\frac{-\epsilon^2}{2\sigma_{\max}^2(\mathbf{a}, \nu) + \frac{2}{3} \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) B_{\mathbf{a}} \epsilon}\right).$$

Next, applying the union bound over all $|C_\nu| = \mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_\infty)$ elements, we obtain:

$$\begin{aligned} P\left(\max_{g' \in C_\nu} |\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| > \epsilon\right) &= P\left(\bigcup_{g' \in C_\nu} \left\{ |\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| > \epsilon \right\}\right) \\ &\leq \sum_{g' \in C_\nu} P\left(|\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| > \epsilon\right) \\ &\leq 2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_\infty) \exp\left(\frac{-\epsilon^2}{2\sigma_{\max}^2(\mathbf{a}, \nu) + \frac{2}{3} \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) B_{\mathbf{a}} \epsilon}\right). \end{aligned}$$

Equating the right-hand side of the union bound to δ , we find the bound ϵ that holds with probability at least $1 - \delta$:

$$\frac{\epsilon^2}{2\sigma_{\max}^2(\mathbf{a}, \nu) + \frac{2}{3} \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) B_{\mathbf{a}} \epsilon} = \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_{\infty})}{\delta} \right).$$

Rearranging terms yields a quadratic equation for ϵ :

$$\underbrace{\epsilon^2 - \left(\frac{2}{3} \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) B_{\mathbf{a}} \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_{\infty})}{\delta} \right) \right) \epsilon}_{b} - \underbrace{\left(2\sigma_{\max}^2(\mathbf{a}, \nu) \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_{\infty})}{\delta} \right) \right)}_c = 0.$$

Using the quadratic formula, the positive root is $\epsilon = \frac{b + \sqrt{b^2 + 4c}}{2}$. By the subadditivity of the square root function ($\sqrt{x + y} \leq \sqrt{x} + \sqrt{y}$ for $x, y \geq 0$), we can simplify this as:

$$\epsilon \leq \frac{b + b + 2\sqrt{c}}{2} = b + \sqrt{c}.$$

Substituting the definitions of b and c back into the inequality, we obtain that with probability at least $1 - \delta$:

$$\max_{g' \in \mathcal{C}_{\nu}} |\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| \leq \sqrt{2\sigma_{\max}^2(\mathbf{a}, \nu) \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_{\infty})}{\delta} \right)} + \frac{2}{3} B_{\mathbf{a}} \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_{\infty})}{\delta} \right).$$

3. Final PAC Bound Derivation Combining this result with the decomposition from Step 1 and Eq. (8), we bound the total excess risk:

$$\begin{aligned} R(\hat{g}) - R(g^*) &\leq 4L_{\mathbf{a}}\nu + 2 \max_{g' \in \mathcal{C}_{\nu}} |\hat{R}_{lin}^{\mathbf{a}}(g') - R(g')| \\ &\leq 4L_{\mathbf{a}}\nu + \sqrt{8\sigma_{\max}^2(\mathbf{a}, \nu) \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_{\infty})}{\delta} \right)} + \frac{4}{3} B_{\mathbf{a}} \max_{m \in [k]} \left(\frac{\theta_m}{n_m} \right) \ln \left(\frac{2\mathcal{N}(\mathcal{G}, \nu, \|\cdot\|_{\infty})}{\delta} \right) \end{aligned}$$

This completes the proof. $\square$

B VARIANCE COMPARISON RESULTS FOR MULTICLASS SETTING

We present the variance comparison results for the multiclass setting. The procedure is analogous to the binary case described in Section 6.1. Since PNU learning is limited to binary classification, we compare only the supervised risk estimator $\hat{R}(g)$ and the proposed linear risk estimator $\hat{R}_{lin}^{\mathbf{a}^*}(g)$.

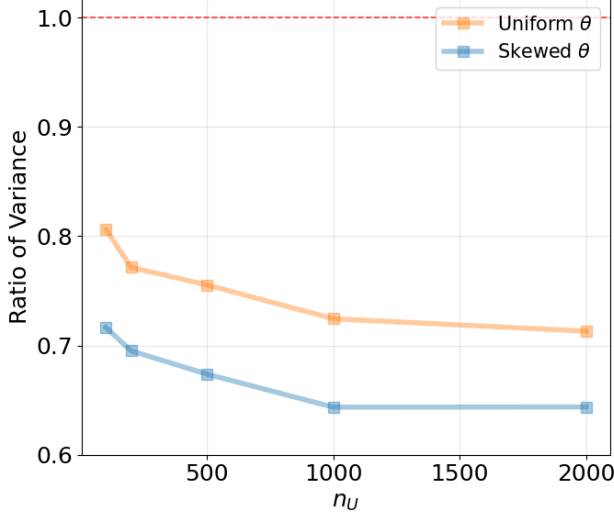
Datasets. We use two multiclass datasets from the UCI Machine Learning Repository: Coverttype and Dry Beans. For each dataset, we use 20% of the data for training and 80% as a held-out pool for evaluation.

Model and training. We train a two-layer neural network with hidden dimension 256 by minimizing the cross-entropy loss with $n_i = 10$ labeled samples per class via the Adam optimizer (learning rate 10^{-3} , weight decay 10^{-4}), with early stopping on validation accuracy.

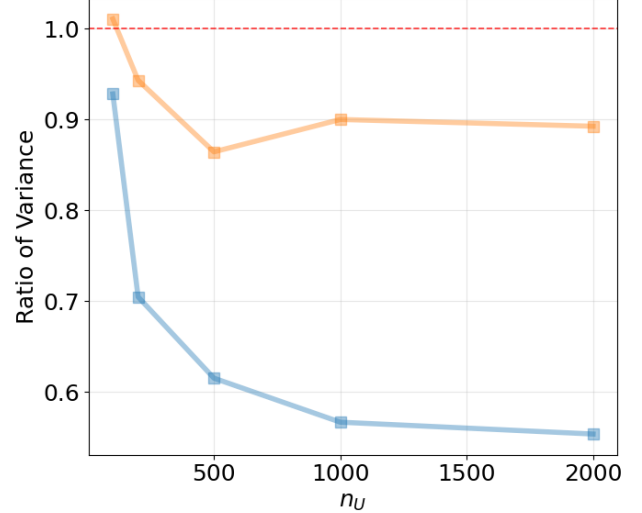
Evaluation loss. We use the cross-entropy (CE) loss $l(g(x), j) = -\log \text{softmax}(g(x))_j$ for the variance comparison.

Evaluation protocol. With the trained classifier g fixed, we evaluate the variance of each risk estimator by repeatedly sampling from the held-out data. The labeled sample sizes for risk estimation are fixed at $n_i = 5$ for all $i \in [k]$, and we vary $n_U \in \{100, 200, 500, 1000, 2000\}$. We consider two class prior settings: (1) *uniform*, where $\boldsymbol{\theta} = (1/7, \dots, 1/7)$, and (2) *skewed*, where $\boldsymbol{\theta} = [57/140, 34/140, 21/140, 13/140, 7/140, 5/140, 3/140] \approx (0.41, 0.24, 0.15, 0.09, 0.05, 0.04, 0.02)$. The optimal parameter $\mathbf{a}^*$ is computed assuming $n_U \rightarrow \infty$ using the covariance matrices estimated from the all evaluation data. For each condition, we conduct 5,000 independent trials to estimate the variance.

As the number of unlabeled samples n_U increases, the variance ratio decreases. The proposed method achieves greater variance reduction under the skewed class prior, consistent with Theorem 4.



(a) Covertypes (CE loss)



(b) Dry Beans (CE loss)

Figure 2: Variance ratio ($\hat{R}_{lin}^{a*} / \hat{R}$) as a function of the unlabeled sample size n_U for uniform and skewed class priors ($k = 7$). The red dashed line indicates the supervised baseline (ratio = 1).

C SSL EXPERIMENTAL DETAILS

We provide full details omitted from the main text for the SSL experiments in Section 6.2.

Validation set. For all settings, the validation set is independently sampled from the original dataset.

Labeled data configurations for multiclass tasks. The per-class labeled sample counts for each regime are as follows.

7-class tasks (Covertypes, Shuttle, Dry Beans):

- Balanced: 10, 20, or 50 per class (totals: 70, 140, 350).
- Mild imbalance: [33, 26, 23, 19, 16, 13, 10] (total 140) and [81, 65, 57, 49, 41, 33, 24] (total 350).
- Severe imbalance: [57, 34, 21, 13, 7, 5, 3] (total 140) and [143, 86, 52, 32, 19, 11, 7] (total 350).

10-class tasks (MNIST):

- Balanced: 20, 50, or 150 per class (totals: 200, 500, 1500).
- Mild imbalance: [102, 82, 71, 61, 51, 41, 31, 26, 20, 15] (total 500) and [306, 245, 214, 184, 153, 122, 92, 77, 61, 46] (total 1500).
- Severe imbalance: [200, 120, 72, 44, 26, 16, 10, 6, 4, 2] (total 500) and [600, 360, 216, 132, 78, 48, 30, 18, 12, 6] (total 1500).

For all multiclass tasks, $n_U = 5,000$.

Model architectures. *MLP (tabular)*: We use an MLP with two hidden layers for all tabular datasets, using ReLU activations and a dropout rate of 0.2. For the Adult, Banknote, Breast Cancer, and Credit Default datasets, we use hidden layer dimensions of [256, 256]. For Covertypes, we use [512, 512], and for Shuttle and Dry Beans, we use [512, 256].

CNN (image, LeNet-like): Here, let $\text{Conv}(c_{in}, c_{out}, k, \text{pad})$ denote a 2D convolutional layer with c_{in} input channels, c_{out} output channels, kernel size $k \times k$, and the specified padding. Let $\text{MaxPool}(k)$ denote a max pooling layer with kernel size $k \times k$. Let $\text{FC}(d_{in}, d_{out})$ denote a fully connected layer with input dimension d_{in} and output dimension d_{out} . For MNIST, the architecture is $\text{Conv}(1, 32, 3, \text{pad}=1) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2) \rightarrow \text{Conv}(32, 64, 3, \text{pad}=1) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2) \rightarrow \text{FC}(64 \times 7 \times 7, 128) \rightarrow \text{ReLU} \rightarrow \text{FC}(128, K)$. For CIFAR-10, the architecture is $\text{Conv}(3, 64, 3, \text{pad}=1) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2) \rightarrow \text{Conv}(64, 128, 3, \text{pad}=1) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2) \rightarrow \text{FC}(128 \times 8 \times 8, 256) \rightarrow \text{ReLU} \rightarrow \text{FC}(256, K)$.

Training protocol. All models are trained with the Adam optimizer (learning rate = 10^{-3} , weight decay 10^{-4}) for up to 200 epochs. Batch sizes are 64 for labeled data and 256 for unlabeled data. Early stopping is based on validation accuracy with patience 20 epochs and minimum improvement threshold 10^{-4} .

Method-specific settings. *PNU* (binary only): The mixing parameter η is computed as $\eta = (\psi_2 - \psi_1)/(\psi_1 + \psi_2)$ where $\psi_m = \theta_m^2/n_m$, following the equal variance assumption [Sakai et al., 2017]. Non-negative risk correction (Δ_j^{nn}) is applied.

Ours (Iter): Training begins with a 20-epoch warm-up period utilizing standard supervised risk prior to joint optimization. The covariance matrices are estimated on the validation split with diagonal shrinkage $\hat{\Sigma}_m \leftarrow (1 - \alpha)\hat{\Sigma}_m + \alpha \text{diag}(\hat{\Sigma}_m)$ ($\alpha = 0.5$), and Tikhonov regularization $\lambda = 10^{-4}$ is added when solving the linear system for the optimal $\mathbf{a}$. Non-negative risk correction (Δ_j^{nn}) is applied.

Ours (EC): The closed-form parameter $a_j = \theta_j(1 - w_j/W)$ with $w_j = \theta_j^2/n_j$ and $W = \sum_m w_m$ is used, requiring no validation data for parameter estimation. The parameter $\mathbf{a}$ is fixed throughout training (no warmup). Non-negative risk correction (Δ_j^{nn}) is applied.

Pseudo Label (PL): The loss is the sum of a standard supervised cross-entropy term and an unsupervised CE term computed from pseudo labels. Initially, a supervised classifier is trained, followed by generating predictions on unlabeled data to create pseudo labels for instances exceeding a specific confidence threshold. The classifier is then retrained using both the original and pseudo labels. The confidence threshold is selected from the set $\{0.8, 0.9, 0.95\}$ for tabular multiclass tasks and $\{0.9, 0.95, 0.99\}$ for image tasks.

VAT: Supervised cross-entropy plus KL-divergence-based local distributional smoothness regularization with the consistency regularization weight is $\lambda_U = 1.0$, the finite-difference step size for approximating the adversarial direction is $\xi = 10^{-6}$, and 1 power-iteration step is used. Batch-normalization running statistics are frozen during the adversarial perturbation computation. The perturbation radius ϵ is selected from $\{0.2, 0.5, 1.0, 2.0\}$ for tabular tasks and $\{0.5, 1.0, 2.0, 4.0\}$ for image tasks.

D ADDITIONAL EXPERIMENTAL RESULTS

This section provides the experimental details and ablation studies that were omitted from the main text because of space constraints. Unless otherwise stated, all table entries report mean $\pm$ std over 30 random seeds. The section is organized as follows. Section D.1 reports results for FixMatch, a recent semi-supervised learning baseline. Section D.2 studies how the number of examples used for covariance estimation affects our method. Section D.3 reports diagnostics for the equal-covariance approximation used by the proposed estimators. Finally, Section D.4 studies the effect of the non-negative risk correction.

D.1 RESULTS FOR A RECENT SSL BASELINE

We report additional results for FixMatch [Sohn et al., 2020], a representative recent SSL method based on pseudo-labeling and consistency regularization. For each unlabeled example, FixMatch first predicts a pseudo-label from a weakly augmented view. It then applies an unsupervised cross-entropy loss to a strongly augmented view of the same example, but only when the maximum predicted class probability exceeds a confidence threshold. Because this threshold can substantially affect performance, we sweep the threshold over $\{0.8, 0.9, 0.95\}$. For each task and labeled-data configuration, we select the threshold that gives the highest mean best-validation accuracy. The selected threshold, the corresponding test accuracy, and the best-validation accuracy are reported in Table 3. The unlabeled-loss weight is fixed at $\lambda_U = 1.0$.

For tabular datasets, the weak view is generated by adding Gaussian noise with $\sigma = 0.1$. The strong view uses Gaussian noise with $\sigma = 0.2$ and feature dropout with drop probability $p = 0.1$. For MNIST-family image tasks, including the binary 4-vs-9 and multiclass MNIST settings, the weak view uses a small random affine transform, degrees = 5 and translation (0.05, 0.05), followed by normalization. The strong view uses a larger random affine transform, degrees = 15, translation (0.1, 0.1), and scale range (0.9, 1.1), followed by random erasing with probability 0.5 and erased-area scale (0.02, 0.1), then normalization. For CIFAR-style image tasks, the weak view uses random horizontal flip and random crop with padding 4; the strong view additionally applies color jitter with brightness/contrast/saturation 0.4 and hue 0.1, random grayscale with probability 0.1, and random erasing with probability 0.5.

Table 3: Classification results for FixMatch. Results are test accuracy and best-validation accuracy in %, reported as mean $\pm$ std over 30 seeds. For binary tasks, Bal. and Imb. denote balanced and imbalanced labeled data setting used in the section 6.2. For MNIST, Bal., Mild, and Sev. follow the regime notation used in the main text.

Task	Regime (n)	Threshold	Test acc. (%)	Best val. acc. (%)
Adult	Bal. (60)	0.95	77.96 $\pm$ 2.11	78.61 $\pm$ 4.04
Adult	Bal. (200)	0.9	78.74 $\pm$ 2.03	78.61 $\pm$ 2.97
Adult	Imb. (60)	0.95	78.97 $\pm$ 2.45	78.67 $\pm$ 3.70
Adult	Imb. (200)	0.9	79.49 $\pm$ 2.42	79.83 $\pm$ 3.20
Banknote	Bal. (60)	0.9	94.88 $\pm$ 3.20	95.55 $\pm$ 3.43
Banknote	Bal. (200)	0.9	97.12 $\pm$ 1.95	97.61 $\pm$ 2.26
Banknote	Imb. (60)	0.95	93.52 $\pm$ 4.84	94.72 $\pm$ 3.16
Banknote	Imb. (200)	0.95	96.46 $\pm$ 2.30	97.22 $\pm$ 2.07
MNIST 4v9	Bal. (60)	0.8	98.83 $\pm$ 0.57	99.68 $\pm$ 0.48
MNIST 4v9	Bal. (200)	0.95	98.99 $\pm$ 0.37	99.68 $\pm$ 0.40
MNIST 4v9	Imb. (60)	0.8	98.89 $\pm$ 0.55	99.63 $\pm$ 0.47
MNIST 4v9	Imb. (200)	0.8	99.03 $\pm$ 0.32	99.67 $\pm$ 0.48
Breast Cancer	Bal. (60)	0.8	93.22 $\pm$ 2.70	96.33 $\pm$ 2.37
Breast Cancer	Bal. (200)	0.9	95.09 $\pm$ 1.99	98.33 $\pm$ 1.91
Breast Cancer	Imb. (60)	0.9	93.39 $\pm$ 2.69	97.00 $\pm$ 2.64
Breast Cancer	Imb. (200)	0.95	95.00 $\pm$ 2.09	98.89 $\pm$ 2.02
Credit	Bal. (60)	0.95	66.97 $\pm$ 7.04	68.17 $\pm$ 8.91
Credit	Bal. (200)	0.95	70.18 $\pm$ 7.01	71.72 $\pm$ 8.61
Credit	Imb. (60)	0.95	72.89 $\pm$ 4.74	76.44 $\pm$ 4.63
Credit	Imb. (200)	0.8	72.22 $\pm$ 7.02	73.72 $\pm$ 8.22
MNIST	Bal. (200)	0.8	97.12 $\pm$ 0.54	98.00 $\pm$ 0.75
MNIST	Bal. (500)	0.95	96.82 $\pm$ 0.76	97.80 $\pm$ 0.98
MNIST	Bal. (1500)	0.8	97.65 $\pm$ 0.55	97.95 $\pm$ 1.07
MNIST	Mild (500)	0.9	96.95 $\pm$ 1.24	97.95 $\pm$ 0.80
MNIST	Mild (1500)	0.8	97.60 $\pm$ 0.53	98.10 $\pm$ 0.81
MNIST	Sev. (500)	0.95	94.71 $\pm$ 3.97	95.35 $\pm$ 4.47
MNIST	Sev. (1500)	0.9	97.39 $\pm$ 0.50	98.00 $\pm$ 0.94

D.2 COVARIANCE-SET SIZE EFFECT

Our method uses a small held-out set to estimate covariance-related quantities. This ablation examines how sensitive **Ours (Iter)** is to the number of held-out covariance-estimation examples per class. We run the experiment on MNIST under two labeled-data regimes: a balanced regime and a severe-imbalance regime. Table 4 shows that performance is fairly stable once a modest number of covariance-estimation examples is available. In particular, performance does not collapse even with very small covariance sets, likely helped by diagonal shrinkage and Tikhonov regularization, and largely saturates around 20–50 examples per class.

D.3 EQUAL-COVARIANCE DIAGNOSTIC

The proposed data-free method relies on an equal-covariance approximation across classes. This diagnostic quantifies how closely the class-conditional covariance matrices agree with one another. In Table 5, Cov. rel. Fro. denotes the average pairwise relative Frobenius distance between class covariance matrices:

$$\frac{\|\Sigma_i - \Sigma_j\|_F}{(\|\Sigma_i\|_F + \|\Sigma_j\|_F)/2}.$$

Smaller values of this measure indicate that the class-conditional covariance matrices are closer to the equal-covariance approximation. These distances are not exactly zero; hence, equal covariance (EC) should be viewed as a data-free approximation, analogous to the equal-variance simplification used for binary PNU. Nevertheless, our experimental results do not show a clear correlation between the degree of EC violation and performance. One possible reason is that this diagnostic measures covariance discrepancies at fixed trained models, whereas Theorem 10 suggests the performance is

Table 4: MNIST covariance-set size ablation using **Ours (Iter)**. Results are test accuracy in %, reported as mean $\pm$ std over 30 seeds.

Regime	Cov. examples per class	Test acc. (%)
Bal. (500)	1	90.96 $\pm$ 1.08
Bal. (500)	2	91.11 $\pm$ 2.11
Bal. (500)	3	91.58 $\pm$ 1.43
Bal. (500)	5	91.54 $\pm$ 1.36
Bal. (500)	10	91.95 $\pm$ 0.92
Bal. (500)	20	92.18 $\pm$ 1.08
Bal. (500)	50	92.14 $\pm$ 0.70
Bal. (500)	100	92.28 $\pm$ 0.84
Bal. (500)	1000	92.55 $\pm$ 0.67
Sev. (500)	1	80.90 $\pm$ 2.23
Sev. (500)	2	78.57 $\pm$ 4.85
Sev. (500)	3	79.57 $\pm$ 3.23
Sev. (500)	5	80.88 $\pm$ 3.14
Sev. (500)	10	81.17 $\pm$ 2.76
Sev. (500)	20	81.75 $\pm$ 2.26
Sev. (500)	50	82.56 $\pm$ 2.40
Sev. (500)	100	81.40 $\pm$ 2.06
Sev. (500)	1000	82.23 $\pm$ 2.61

affected by the maximum variance over the hypothesis class. This distinction may explain why this local measure of EC violation is not directly predictive of empirical performance.

Table 5: Equal-covariance diagnostic summary. Results are mean $\pm$ std over 30 seeds. Smaller Cov. rel. Fro. values indicate closer agreement with the equal-covariance approximation.

Setting	Classes	Test acc. (%)	Cov. rel. Fro.
Covertypes, Bal. (140)	7	56.58 $\pm$ 2.28	1.5110 $\pm$ 0.0864
Covertypes, Mild (350)	7	63.71 $\pm$ 1.51	1.5344 $\pm$ 0.0847
Covertypes, Sev. (350)	7	64.26 $\pm$ 1.71	1.5386 $\pm$ 0.0794
Dry Beans, Bal. (350)	7	91.56 $\pm$ 0.71	1.1433 $\pm$ 0.0275
Dry Beans, Sev. (140)	7	85.28 $\pm$ 1.94	1.1341 $\pm$ 0.1139
MNIST, Mild (500)	10	89.74 $\pm$ 0.99	1.1260 $\pm$ 0.0847
MNIST, Sev. (500)	10	80.94 $\pm$ 2.98	1.0817 $\pm$ 0.0701
Shuttle, Bal. (70)	7	96.74 $\pm$ 1.57	1.3476 $\pm$ 0.0407
Shuttle, Mild (140)	7	97.88 $\pm$ 0.82	1.3545 $\pm$ 0.0314

D.4 NON-NEGATIVE RISK-CORRECTION ABLATION

We study the effect of the non-negative risk correction introduced in Section 5.2. Tables 6–10 compare **Ours (Iter)** with and without this correction. In the tables, “NN on” is the default method with the non-negative risk correction, and “NN off” is the same method without it. We also report

$$\Delta = \text{accuracy}(\text{NN off}) - \text{accuracy}(\text{NN on})$$

in percentage points. A negative Δ means that the correction improves accuracy, while a positive Δ means that removing the correction gives higher accuracy. Overall, the correction has a small effect on many tabular datasets. However, it is highly effective on image datasets with larger accuracy improvements.

Table 6: Non-negative risk-correction ablation on binary tabular tasks. Results are test accuracy in %, reported as mean $\pm$ std over 30 seeds.

Task	Labeled	NN on (%)	NN off (%)	Δ (pp)
Adult	(15, 45)	80.3 $\pm$ 1.6	80.2 $\pm$ 1.6	-0.1
Adult	(30, 30)	80.8 $\pm$ 1.5	80.9 $\pm$ 1.1	+0.1
Adult	(50, 150)	82.2 $\pm$ 1.0	82.1 $\pm$ 0.9	-0.1
Adult	(100, 100)	82.4 $\pm$ 1.0	82.5 $\pm$ 1.0	+0.1
Banknote	(15, 45)	97.3 $\pm$ 1.5	96.9 $\pm$ 2.0	-0.4
Banknote	(30, 30)	97.8 $\pm$ 1.1	97.9 $\pm$ 1.4	+0.1
Banknote	(50, 150)	98.6 $\pm$ 0.9	98.6 $\pm$ 0.9	0.0
Banknote	(100, 100)	98.8 $\pm$ 1.0	98.8 $\pm$ 1.0	0.0
Breast Cancer	(15, 45)	94.5 $\pm$ 2.3	94.6 $\pm$ 2.4	+0.2
Breast Cancer	(30, 30)	94.4 $\pm$ 1.8	94.5 $\pm$ 2.0	+0.2
Breast Cancer	(50, 150)	95.5 $\pm$ 1.7	95.6 $\pm$ 1.7	0.0
Breast Cancer	(100, 100)	95.6 $\pm$ 1.8	95.6 $\pm$ 1.7	+0.1
Credit	(15, 45)	78.2 $\pm$ 2.0	78.3 $\pm$ 1.8	+0.1
Credit	(30, 30)	78.9 $\pm$ 1.2	78.9 $\pm$ 1.2	0.0
Credit	(50, 150)	78.9 $\pm$ 1.3	79.0 $\pm$ 1.2	0.0
Credit	(100, 100)	79.3 $\pm$ 1.1	79.3 $\pm$ 1.1	0.0

Table 7: Non-negative risk-correction ablation on binary image tasks. Results are test accuracy in %, reported as mean $\pm$ std over 30 seeds.

Task	Labeled	NN on (%)	NN off (%)	Δ (pp)
MNIST 4v9	(15, 45)	94.7 $\pm$ 1.9	92.3 $\pm$ 3.4	-2.5
MNIST 4v9	(30, 30)	95.6 $\pm$ 1.6	94.6 $\pm$ 3.1	-1.0
MNIST 4v9	(50, 150)	97.3 $\pm$ 0.9	96.9 $\pm$ 1.1	-0.4
MNIST 4v9	(100, 100)	97.7 $\pm$ 0.7	97.4 $\pm$ 1.1	-0.3
CIFAR CvD	(15, 45)	55.4 $\pm$ 1.6	55.8 $\pm$ 2.1	+0.4
CIFAR CvD	(30, 30)	55.8 $\pm$ 2.0	55.2 $\pm$ 2.1	-0.6
CIFAR CvD	(50, 150)	58.2 $\pm$ 2.1	58.7 $\pm$ 1.6	+0.5
CIFAR CvD	(100, 100)	59.6 $\pm$ 1.7	58.7 $\pm$ 1.7	-0.9

Table 8: Non-negative risk-correction ablation on multiclass tasks with equal labeled samples per class. Results are test accuracy in %, reported as mean $\pm$ std over 30 seeds.

Task	Labeled	NN on (%)	NN off (%)	Δ (pp)
MNIST	20	86.9 $\pm$ 1.3	85.6 $\pm$ 1.8	-1.3
MNIST	50	91.8 $\pm$ 0.5	91.4 $\pm$ 1.3	-0.4
MNIST	150	95.4 $\pm$ 0.4	95.3 $\pm$ 0.6	-0.1
CIFAR-10	20	31.7 $\pm$ 1.4	28.9 $\pm$ 2.1	-2.9
CIFAR-10	50	38.4 $\pm$ 1.2	35.0 $\pm$ 2.0	-3.3
CIFAR-10	150	47.1 $\pm$ 1.1	44.3 $\pm$ 1.2	-2.7
Coverttype	10	52.0 $\pm$ 2.9	53.1 $\pm$ 3.1	+1.2
Coverttype	20	55.8 $\pm$ 1.3	56.3 $\pm$ 2.8	+0.6
Coverttype	50	61.2 $\pm$ 2.1	61.4 $\pm$ 1.9	+0.2
Shuttle	10	96.7 $\pm$ 1.2	96.6 $\pm$ 1.5	-0.1
Shuttle	20	97.3 $\pm$ 1.0	97.5 $\pm$ 0.9	+0.2
Shuttle	50	97.3 $\pm$ 1.0	97.8 $\pm$ 0.9	+0.5
Dry Beans	10	87.6 $\pm$ 1.6	87.5 $\pm$ 1.7	-0.1
Dry Beans	20	89.9 $\pm$ 0.9	89.6 $\pm$ 1.1	-0.3
Dry Beans	50	91.3 $\pm$ 0.6	91.3 $\pm$ 0.6	0.0

Table 9: Non-negative risk-correction ablation on multiclass tasks with mild imbalance. Results are test accuracy in %, reported as mean $\pm$ std over 30 seeds.

Task	Labeled	NN on (%)	NN off (%)	Δ (pp)
MNIST	500	89.8 $\pm$ 1.4	89.3 $\pm$ 1.8	-0.5
MNIST	1500	94.3 $\pm$ 0.8	94.0 $\pm$ 0.9	-0.3
CIFAR-10	500	34.7 $\pm$ 1.0	33.2 $\pm$ 1.9	-1.5
CIFAR-10	1500	42.6 $\pm$ 1.0	42.2 $\pm$ 2.1	-0.4
Covertypes	140	57.4 $\pm$ 2.1	58.5 $\pm$ 2.4	+1.0
Covertypes	350	62.7 $\pm$ 1.4	62.5 $\pm$ 1.1	-0.2
Shuttle	140	96.9 $\pm$ 1.0	97.3 $\pm$ 1.5	+0.4
Shuttle	350	97.6 $\pm$ 1.1	97.7 $\pm$ 1.0	+0.1
Dry Beans	140	89.1 $\pm$ 1.7	88.9 $\pm$ 1.8	-0.2
Dry Beans	350	90.7 $\pm$ 0.9	90.6 $\pm$ 1.0	-0.1

Table 10: Non-negative risk-correction ablation on multiclass tasks with severe imbalance. Results are test accuracy in %, reported as mean $\pm$ std over 30 seeds.

Task	Labeled	NN on (%)	NN off (%)	Δ (pp)
MNIST	500	81.1 $\pm$ 1.1	74.9 $\pm$ 3.1	-6.2
MNIST	1500	89.5 $\pm$ 1.7	86.9 $\pm$ 2.3	-2.6
CIFAR-10	500	28.1 $\pm$ 1.3	25.6 $\pm$ 2.1	-2.5
CIFAR-10	1500	34.1 $\pm$ 1.9	33.4 $\pm$ 1.8	-0.7
Covertypes	140	58.0 $\pm$ 1.7	59.0 $\pm$ 2.2	+0.9
Covertypes	350	63.3 $\pm$ 1.7	63.4 $\pm$ 1.6	+0.1
Shuttle	140	97.2 $\pm$ 0.9	97.3 $\pm$ 1.1	+0.1
Shuttle	350	97.6 $\pm$ 0.8	97.6 $\pm$ 0.8	+0.1
Dry Beans	140	84.9 $\pm$ 2.6	85.5 $\pm$ 2.4	+0.6
Dry Beans	350	89.1 $\pm$ 1.5	88.9 $\pm$ 1.5	-0.1